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Abstract

A recent review found that 11% of published factor models are hierarchical with second-order factors. However, dedicated recommendations for evaluating hierarchical model fit have yet to emerge. Traditional benchmarks like $RMSEA < 0.06$ or $CFI > 0.95$ are often consulted, but they were never intended to generalize to hierarchical models. Through simulation, we show that traditional benchmarks perform poorly at identifying misspecification in hierarchical models. This corroborates previous studies showing that traditional benchmarks do not maintain optimal sensitivity to misspecification as model characteristics deviate from those used to derive the benchmarks. Instead, we propose a hierarchical extension to the dynamic fit index (DFI) framework, which automates custom simulations to derive cutoffs with optimal sensitivity for specific model characteristics. In simulations to evaluate performance, results showed that the hierarchical DFI extension routinely exceeded 95% classification accuracy and 90% sensitivity to misspecification whereas traditional benchmarks rarely exceeded 50% classification accuracy and 20% sensitivity.

Dynamic Fit Index Cutoffs for Hierarchical and Second-Order Factor Models

The main interest of this paper is evaluating the fit of hierarchical (a.k.a. second-order or higher-order) factor models (e.g., Bollen, 1989, p. 313). In these models, there are *first-order* factors that load directly onto observed variables (similar to a standard confirmatory factor model) and there are one or more *second-order* factors that load onto the first-order factors (and indirectly onto the items). A hierarchical factor model is technically a structural equation model rather than a confirmatory factor analysis (Brown, 2015, p. 321)—the first-order factors form the measurement model (i.e., structuring the covariance among observed item responses) while the second-order factors form the structural model for the covariance among the first-order factors (Kline, 2010, p. 249). Although hierarchical factor models can technically be expressed as a special case of a bifactor model with proportionality constraints (Yung et al., 1999), we narrow our focus to hierarchical factor models and do not consider generalizations to the broader bifactor context.

As an overview of this paper, we first discuss approximate fit indices for latent variable models and the origin of the common Hu and Bentler (1999; hereafter denoted as HB) cutoffs for indices like RMSEA, CFI, and SRMR. We then discuss how these cutoffs were derived from a simulation and how methodological research has shown that traditional cutoffs do not generalize to models outside of those studied by HB (e.g., Fan & Sivo, 2007; Greiff & Heene, 2017; Nye & Drasgow, 2011). Specifically, HB studied a 3-factor confirmatory factor model and their results therefore do not necessarily generalize to hierarchical factor models.

This is a prevalent problem because a review by Jackson et al. (2009) reports that 11% of factor models in psychology studies are hierarchical factor models, but no dedicated guidance on fit index interpretation exists for researchers working with these models. In absence of such dedicated guidelines, didactic resources for hierarchical factor models often recommend generalizing cutoffs from HB (e.g., Caspi et al., 2014, p. 124; Murray & Johnson, 2013, p. 411; though see Brunner et al., 2012 for a didactic resource where this ambiguity is discussed), even

though they were never intended to be generalized in this way. For instance, Hu and Bentler (1998) state, “our findings suggest that the performance of fit indices is complex and that additional research with a wider class of models and conditions is needed” (p. 446) and continue by writing, “Further work should be performed to explore the limits of generalizability in various ways, for example, across types of structural models and overparameterized models” (p. 450).

The main goal of this paper is to provide researchers with a more refined set of fit index cutoffs for hierarchical factor models within the dynamic fit index (DFI) framework (McNeish & Wolf, 2021). DFI is an automated version of ideas proposed by Millsap (2007, 2013) to perform custom simulations that derive fit index cutoffs tailored to the specific model being evaluated. Customization limits concerns about generalizing cutoffs derived from other conditions and maintains optimal classification properties of cutoffs regardless of the model’s characteristics.

The DFI framework currently encompasses one-factor and multiple-factor confirmatory factor analysis (CFA) models, but it has not been extended to hierarchical factor models despite their widespread use in fields such as personality (e.g., Brunner et al., 2012; DeYoung, 2006; Digman, 1997; Krueger et al., 2007), psychopathology (Caspi et al., 2014; Markon et al., 2005; Wright et al., 2010; Zinbarg, Barlow, & Brown, 1997), education (e.g., Marsh & Craven, 2006; Marsh & Hocevar, 1985; Martin, 2007; Rijmen, 2010), and intelligence and creativity (e.g., Carroll, 1993; McGrew, 2009; Silvia, 2008).

Researchers fitting hierarchical factor models have noted a desire to implement the DFI approach but have lamented that it does not yet exist. For instance, Johnson et al. (2023) write, “McNeish and Wolf (2021) developed a simulation-based tool to dynamically adjust fit index cutoff values (for adequate model fit) based on a model’s standardized loadings, degrees of freedom, and sample size. However, the tool does not yet provide estimates for higher-order models, like the one in the current study” (pp. 13-14). We therefore propose an extension of DFI for hierarchical factor models to broaden the types of models to which this method can be applied, with an explicit interest in helping researchers evaluate their hierarchical factor models

as accurately as possible. We consider an empirical example to demonstrate potential weaknesses of applying traditional cutoffs to hierarchical factor models. We then conduct simulation studies to evaluate the performance of the proposed extension of DFI to hierarchical factor models. Performance of the proposed hierarchical extension of DFI is compared to traditional cutoffs to assess the extent to which this method improves upon the status quo. Limitations and extensions are then discussed.

Background

Evaluating fit is a central step when applying latent variable models to empirical data and there are two main types of fit metrics applied to assess global fit of a model. First, tests of *exact* fit operate in the null hypothesis significance testing framework and inferentially assess the null hypothesis that the model-implied covariance matrix is equal to the covariance matrix of the observed data. The most commonly reported exact fit statistic is the maximum likelihood test statistic (T_{ML}), which is colloquially referred to as “the χ^2 test” because T_{ML} asymptotically follows a χ^2 distribution, assuming normality. Tests of exact fit are praised for having a clear null hypothesis and providing definitive conclusions (e.g., Barrett, 2007). However, they are also criticized because exact fit is not necessarily required for a model to have utility (Browne & Cudeck, 1992; Meehl, 1978; Mulaik, 2009) and a rejection of the null hypothesis of exact fit may be triggered by small misspecifications or modest deviations from normality when sample sizes are large (Hu et al., 1992; Saris et al., 1987). Browne and Cudeck (1993) capture this sentiment directly by saying, “in applications of the analysis of covariance structures in the social sciences it is implausible that any model that we use is anything more than an approximation to reality. Since a null hypothesis that a model fits exactly in some population is known a priori to be false, it seems pointless even to try and test whether it is true” (p. 137).

As an alternative, *approximate* fit indices like RMSEA, CFI, and SRMR were developed. Whereas exact fit tests are interested in the *presence* of misspecification, approximate fit indices are interested in the *magnitude* of misspecification. In this way, fit indices operate more like

effect sizes where the goal is to quantify the degree of misspecification in the model (e.g., Ropovik, 2015). This allows researchers to supplement binary decisions from exact fit tests and evaluate fit more holistically. However, these indices do not always have an intuitive scale (e.g., RMSEA is an unstandardized effect size; Maydeu-Olivares et al., 2018) and there is uncertainty regarding which values denote that misspecification is practically large enough to cause concern. That is, the prevailing practical question is often, “how much misspecification is too much misspecification?” (Tanaka, 1987).

When approximate fit indices were first introduced, values interpreted as evidence of reasonable fit were mostly based on heuristics or personal experience (Marsh et al., 2004; West et al., 2023). For instance, when discussing which values of RMSEA are reasonable, Browne & Cudeck (1993) suggested, “practical experience has made us feel that a value of the RMSEA of about 0.05 or less would indicate a close fit of the model in relation to the degrees of freedom ... We are also of the opinion that a value of about 0.08 or less ... would indicate a reasonable error of approximation and would not want to employ a model with an RMSEA greater than 0.1” (p. 144). Similarly, when discussing reasonable values for CFI, Bentler and Bonett (1980) wrote, “experience will be required to establish values of the indices that are associated with various degrees of meaningfulness of results. In our experience, models with over fit indices of less than .9 can usually be improved substantially” (p. 600).

HB sought to introduce systematicity to fit index interpretation and assess the performance of these heuristic suggestions. They conducted a simulation study to empirically determine which fit index values could consistently detect misspecification in CFA. They simulated data and then fit CFA models that were either (a) known to be correct or (b) known to be misspecified. They recorded values of fit indices and determined which values of which indices could consistently differentiate between correct and misspecified models. This simulation is the origin of the now ubiquitous $RMSEA \leq 0.06$, $SRMR \leq 0.08$, and $CFI \geq 0.95$, fit index cutoffs which remain popular today (Jackson, 2009; Zyphur et al., 2023). These cutoffs

have had immense influence on behavioral research as evidenced by HB receiving over 100,000 citations. HB transformed how researchers approach fit evaluation because cutoffs were determined empirically rather than through heuristics.

To demonstrate how this process works, we recreated HB's simulation for RMSEA, which produced the plot shown in the top panel of Figure 1. The dark grey distribution represents fit indices when the model is known to be correct, and the light grey distribution represents fit indices when the model is known to be misspecified. In HB's simulation, the "misspecified" model omitted one 0.50 standardized cross-loading in a 3-factor model with 15 items. In these conditions, an RMSEA value of 0.06 would correctly reject most models from the light grey distribution corresponding to misspecified models while also correctly retaining most models from the dark grey distribution corresponding to correct models (i.e., nearly all of the light grey distribution is to the right of the 0.06 cutoff where all of the dark grey distribution is to the left of the cutoff). The logic is therefore that models with RMSEA values of 0.06 are consistent with non-ignorable misspecifications whereas models with RMSEA values below 0.06 have no or ignorable misspecifications. Cutoffs for other fit indices were derived in a similar manner by determining which fit index values would optimally differentiate between models that were known to be correct or known to be misspecified based on the simulated data.

HB was instrumental in making approximate fit evaluation more objective. Nonetheless, there are important limitations with these traditional cutoffs, which Hu and Bentler themselves have noted (e.g., Hu & Bentler, 1998, p. 446; Hu & Bentler, 1999, p. 27). Primarily, the cutoffs were derived from simulation and therefore only generalize to the conditions studied by HB (Paxton et al., 2001), which include CFA models with 3 factors, 15 items, and standardized loadings between 0.70 and 0.80 for items with no cross-loadings. Recent studies have shown that the traditional cutoffs are not stable and that applying the same principles under different conditions (e.g., number of factors, number of items, standardized loading magnitude, model type) leads to different cutoffs (e.g., Browne et al., 2002; Hancock & Mueller, 2011; Heene et al.,

2011; McNeish et al., 2018; Miles & Shevlin, 2007; Saris et al., 2009; Shi et al., 2019; Savalei, 2012; Sivo et al., 2006; Tomarken & Waller, 2003). An often-overlooked paper by Schönemann (1981) presented a proof anticipating these issues around the time that fit indices were just beginning to receive attention.

In terms of Figure 1, this means that the mean and variance of the two distributions shift under different conditions. For instance, the bottom panel in Figure 1 shows RMSEA distributions using HB's same simulation design with an identical misspecification but changing the loadings to 0.50-0.60 instead of 0.70-0.80. The distributions shift to the left considerably and the traditional 0.06 RMSEA cutoff now does a poor job identifying misspecified models. Essentially, fit index values that optimally differentiate between the two distributions are not fixed, but change based on the characteristics of the model. In other words, fit indices encode misspecification differently depending on the characteristics of the model, so an RMSEA of 0.06 does not necessarily convey identical magnitude of misspecification in all models.

The reason for this fluctuation is that HB is essentially a power analysis (McNeish & Wolf, 2021). In a traditional power analysis, the goal is to determine the sample size at which statistical tests are sensitive to the presence of an effect. The goal of the HB simulation was to determine fit index values that are sensitive to the presence of model misspecification. The general goal of both is the same – to determine some value of a target quantity that optimizes sensitivity to the presence of some phenomena. In power analysis, the target quantity is sample size and the phenomenon is usually the effect of a predictor or group assignment. In HB, the target quantity is a fit index and the phenomenon is a model misspecification. Just as sufficient sample size (the target quantity) with optimal sensitivity to a non-null effect in a traditional power analysis is affected by model characteristics (e.g., number of predictors, strength of relations between variables), so too are values of fit indices (the target quantity) with optimal sensitivity to misspecification.

In power analysis, there are no fixed sufficient sample sizes that indiscriminately apply

to all studies. For instance, sufficient sample size for a study using a within-subjects design will be different compared to a between-subjects design – there is no single sample size like $N = 100$ that is considered sufficient for all studies. Similarly, whereas an RMSEA of 0.06 might be indicative of good fit under the conditions studied by HB, the same 0.06 RMSEA might indicate poor fit – for instance – if the standardized loadings are 0.50-0.60 as in the bottom panel of Figure 1. Just as power analyses are conducted on a case-by-case basis to isolate which sample size is optimally sensitive to the presence of a non-null effect, some have argued that fit index cutoffs should similarly be derived for each model to render more accurate conclusions about model fit (Kim & Millsap, 2014; Millsap, 2007, 2013). As is the focus of this paper, fit index cutoffs derived from a 3-factor model will not necessarily generalize to hierarchical factor models and cutoffs dedicated to nuances of hierarchical models are needed.

Empirical Example

As one example, consider the hierarchical factor model fit by Reynolds and Keith (2017) to 16 cognitive ability items from the Weschler Intelligence Scale for Children - Fifth Edition (WISC-V) administered to 2200 children between the ages of 6 and 16. The path diagram with standardized estimates for this model is shown in Figure 2. This model featured five first-order factors (Processing Speed, Working Memory, Fluid Reasoning, Visual Spatial, and Verbal Comprehension) and one general second-order factor (General Intelligence). Each first-order factor had between 2 and 4 items. Item 7 has a cross-loading and there was a theoretical disturbance correlation between the Fluid Reasoning and Visual Spatial first-order factors. All items generally loaded strongly on their respective factor with standardized loadings ranging from 0.41 to 0.85. The loadings from the general second-order factor to the first-orders factors were also generally strong with values between 0.51 and 0.97. The hierarchical model did not fit exactly ($\chi^2(97) = 329.17, p < .01$) but RMSEA = 0.033 and CFI = 0.985, which was interpreted as acceptable fit in accordance with traditional cutoffs suggested by HB.

Nonetheless, the characteristics of this model deviate from those studied in HB. For

instance, this model has 5 factors in the measurement model as opposed to the 3 factors in HB, the loadings in the measurement model are more heterogeneous (range = 0.41-0.85) than the loadings in HB (range = 0.70-0.80), and the (model-implied) factor correlations were also much stronger in this model (range = 0.42-0.84) than in the model studied by HB (range = 0.30-0.50). The biggest difference, of course, is that the model type is different because there is a second-order factor that loads on first-order factors. That is, this model structures the covariances between the first-order factors with a second-order factor as opposed to fitting a CFA that simply estimates the factor covariances. Taken together, it is unclear how well HB cutoffs apply to this model. As emphasized in previous research, it is unclear which values of RMSEA and CFI are sensitive to misspecification in these conditions and whether an RMSEA of 0.033 or a CFI of 0.985 suggest an acceptably small misspecification in a hierarchical factor model.

To be clear, we are not faulting the methodology in Reynolds and Keith (2017). Contrarily, we use this study because it was thorough and transparent in its reporting practices and all relevant information was provided in the analysis and discussion of results so that everything could be reproduced. The authors applied widely accepted methods and there is little more the authors could have realistically done to address ambiguities about fit interpretation raised here. This is the precise motivation for the current paper – the methodological literature has not provided researchers with accessible options to address fit index interpretation issues and potentially suboptimal benchmarks continue to persist in situations where their adequacy is unknown or even known to be poor. In absence of a more appropriate and accessible method, imperfect methods continue to be generalized to contexts where they are likely inappropriate.

The next section discusses DFI as one recent proposal to remedy generalization issues in the context of CFA. We then propose an extension to the case of hierarchical factor models to expand the types of models to which DFI can be applied. We later return to these data and reanalyze them using the hierarchical DFI extension to assess whether conclusions may change

if fit index cutoffs were tailored to the characteristics of this model rather than being generalized from HB's 3-factor CFA simulation.

Improving Upon Traditional Cutoffs

Criticisms of generalizing cutoffs from a single simulation have been noted for nearly 20 years (e.g., going back to Marsh et al., 2004, 2005) but a review of CFA reporting practices by Jackson et al. (2009) noted “we also did not find evidence that warnings about strict adherence to Hu and Bentler’s suggestions were being heeded” (p. 18). A more recent review by Zyphur et al. (2023) notes that HB cutoffs continue to be the most popular method by which fit is evaluated (p. 504). The bigger issue is therefore not to continue pointing out that there are issues with overgeneralizing HB cutoffs – especially without providing empirical researchers with a viable alternative – as this has already been articulately covered many times (e.g., Chen et al., 2008; Hancock & Mueller, 2011; Heene et al., 2011) without a sizeable shift in conventional practices. Instead, a more impactful focus may be to provide researchers with accessible alternatives so that they can heed previous warnings to move past HB cutoffs and adopt more appropriate methods to gauge fit more accurately.

Millsap (2007, 2013) proposed that researchers could conduct customized simulations to derive fit index cutoffs that are optimally sensitive to misfit for the specific model characteristics present in the model being evaluated (see also, Kim & Millsap, 2014; Pornprasertmanit et al., 2013). The idea is that the logic of HB is sound, but the issue is generalizability. If HB were able to obtain reasonable cutoffs for the narrow conditions they studied, then researchers could conduct a simulation with conditions that match their own model’s characteristics to derive customized cutoffs tailored specifically to their model. This maps onto the idea of HB as a power analysis – just like power analyses are conducted uniquely for each set of circumstances to determine sufficient sample sizes under certain conditions, so too must fit index cutoffs be uniquely derived for each unique set of model characteristics.

Millsap’s idea is conceptually alluring but has some challenges for practical

implementation and has therefore not been widely adopted in empirical studies (e.g., McNeish & Wolf, 2021). First, Monte Carlo simulation is not a methodological skill possessed by most researchers (e.g., Arend & Schäfer, 2019; Green & MacLeod, 2016), so many researchers may struggle to apply the method. Second, it takes time to program a simulation from scratch, so even researchers who are knowledgeable about Monte Carlo simulation may perceive such an approach as being prohibitively time consuming. Third, custom simulation requires researchers to identify a meaningful potential misspecification to which the cutoffs should be sensitive. HB considered omitted cross-loadings and omitted factor covariances which could serve as a starting point, but these paths may not exist in all models and alternatives would need to be considered in some cases (e.g., one-factor models cannot have omitted cross-loadings or omitted factor covariances; McNeish & Wolf, 2022). DFI has recently been developed to overcome barriers with customized simulation of fit index cutoffs by making the approach more accessible.

Dynamic Fit Indices

To improve accessibility, DFI automates the two most challenging parts of the custom simulation method. First, DFI uses an algorithm to identify hypothetical misspecifications to the user's model that are of similar magnitude to the misspecifications considered in HB so that researchers do not need to work out the linear algebra themselves. By "user's model", we mean the factor model a researcher is trying to evaluate. Second, DFI writes and executes code for a Monte Carlo simulation so that researchers do not need to manually produce any software code. This lowers the barrier to implementation because researchers with varying quantitative backgrounds can apply the method, which is available through the open-source `dynamic R` package (Wolf & McNeish, 2022) or through a free web-based Shiny application (Wolf & McNeish, 2021). Informally, the goal is for DFI to be the "GPower of model fit" such that a useful but technical idea is simplified for widespread use by researchers with diverse levels of methodological training and backgrounds. Essentially, the motivation behind DFI is that HB cutoffs endure not necessarily because they are the best but because they are the easiest to

apply. Therefore, any alternative to HB must be similarly accessible to affect practice.

Full details of this process for models currently supported by DFI are available from McNeish and Wolf (2021, 2022) and high-level overviews focusing on software implementation are also available in Wolf & McNeish (2022, 2023).¹ As a general overview of how the DFI method works, the basic idea is that the user's model is the basis for the population model in a Monte Carlo simulation. First, 500 datasets are generated from the covariance matrix implied by the user's model and parameter estimates. This ensures that the conditions of the simulation match the characteristics of the model being evaluated (e.g., the number of factors, variables, strength of loadings, sample size). The user's model is then fit to all 500 simulated datasets and fit indices are recorded. This forms the dark grey distribution in Figure 1 where the model is consistent with the data and tailored to the user's model characteristics.

Figure 1 also contains a light grey distribution corresponding to fit indices when the model is misspecified. Therefore, in the next step of DFI, an algorithm searches for possible paths that could have been included in the model but were not (e.g., cross-loadings or residual correlations). It determines which paths and which magnitudes of those paths that – if added – would render the fitted model misspecified at a similar magnitude as models considered in HB. Once these additional paths have been determined, 500 datasets are simulated from the covariance matrix implied by the augmented model consisting of the user's model *plus additional hypothetical paths* (using the same number of factors, items, sample size, etc.). This creates simulated data for which the user's model is misspecified. The user's original model is then fit to these 500 simulated datasets and the fit indices are recorded. This will form the light grey distribution in Figure 1 – tailored to the model's characteristics – because the model is known to be inconsistent with these simulated data. This process can also be repeated with increasingly severe misspecifications to derive cutoffs that are sensitive to different magnitudes

¹ For didactic resources on software implementation of the DFI method, also see the DFI frequently asked questions document (<https://psyarxiv.com/2zk7g/>) or the vignettes located at <https://github.com/melissagwolf/dynamic>

of misspecification. Researchers do not need to alter their models in any way to make them compatible with DFI. If the model being evaluated already has cross-loadings or residual correlations (e.g., to account for order or reverse-coding effects), those paths can remain in the model and fit of models with these paths can still be evaluated with DFI. The intention of DFI is to be practical and accessible without requiring that researchers artificially alter their model; the model can be entered as desired and DFI will operate based on whichever paths are in the model.

Once a distribution of fit indices has been derived assuming the user's model is correct and assuming the user's model is misspecified, the same decision rule shown in Figure 1 can be applied to determine a customized fit index for the user's model. The default decision rule to determine the DFI cutoff – which is adopted from guidelines used in HB – is that 95% of misspecified models should be rejected while not more than 5% of correct models are rejected. If no fit index exists that satisfies these criteria, then alternative decision rules may be applied (e.g., 90% sensitivity to hypothetical misspecifications instead of 95%).

If applying this idea to the bottom panel of Figure 1, the DFI cutoff would be placed around 0.025 because most of the correct models (in dark grey) are below this value while most of the misspecified models (in light grey) are above this value. This is what makes DFI “dynamic” – it adapts the cutoffs to the changing properties of fit index distributions to maintain cutoffs with optimal properties regardless of the model characteristics. Note that it is possible that no DFI cutoff may exist if fit indices are not differentiable between correct and misspecified models, which can occur when the distributions overlap too much. This is more prevalent with smaller samples because sampling variability will increase, making it more likely that distributions overlap (McNeish & Wolf, 2022).

In essence, DFI has the same idea as HB such that there is a correct model and a misspecified model – DFI just works in reverse. HB generated data from a true model and then fit a model they knew to be misspecified. DFI is based on the user's model in order to tailor the

simulation to the exact characteristics of the model being evaluated. However, a consequence of basing the simulation on a real model fit to real data is that the truth is unknown. To remedy this, DFI works in reverse by treating the user’s model as hypothetically misspecified and trying to find a hypothetical “true” model from which to generate data that would render the user’s model similarly misspecified as models used in HB. This does not imply that these hypothetical misspecifications are actually present nor is DFI trying to locate specific misspecifications in the user’s model. Hypothetical misspecifications are just used to determine the scaling of fit indices under particular conditions and to better understand how fit indices behave for the user’s model characteristics. In the end, DFI is basically a generalization of HB’s simulation – but rather than producing fixed cutoffs from fixed simulation conditions, the simulation conditions in DFI actively adapt to the model being evaluated to produce dynamic cutoffs tailored to the user’s specific model characteristics.

Additional Proposed Methods

Although our focus is on DFI, there are related methods that have similar motivations. Pornprasertmanit et al. (2013) was one of the first attempts to create a simulation-based method designed for evaluating fit and was accompanied by functions in the `simsem` R package to facilitate implementation (Pornprasertmanit et al., 2022). This method successfully automates execution of the simulation design, but researchers must either specify the hypothetical misspecification they wish to test or choose from preselected options, which may limit its accessibility to empirical researchers. The main differentiation between DFI and the method implemented in `simsem` is that DFI attempts to determine a relevant misspecification to test to increase accessibility of simulation-based methods.

Groskurth et al. (2022) proposed a simulation-based approach that uses a ROC curve to balance sensitivity and specificity of classifications between fit index distributions assuming the user’s fitted model is correct and assuming the user’s fitted model is misspecified. Similar to the approach implemented in `simsem`, this approach deviates from DFI in that researchers must

identify the misspecification to be tested whereas DFI has an internal process to identify misspecifications to test in attempt to make the method more accessible. DFI also prioritizes cutoffs that will not erroneously reject acceptable models even if this lowers sensitivity to misspecification whereas the ROC approach seeks to produce cutoffs that balance error rates (which may include higher chances of rejecting acceptable models). As of this writing, the method can be applied through a Shiny application (<https://kg11.shinyapps.io/tailoredcutoffs/>) but there is not yet a corresponding R package.

The flexible cutoff (FCO) approach is another option that simulates custom fit index cutoffs (Niemand & Mai, 2018). In FCO, the population model from which many datasets are generated is based on the user's model. The user's model is subsequently fit to each of these generated datasets. The result is a distribution of fit index cutoffs assuming that the user's model is correct (i.e., the data are generated from the same model that is eventually fit to the data). The cutoff is then based upon the extreme tail of the simulated fit index distribution. For instance, for a smaller-is-better index like RMSEA, the 95th percentile could be used such that any model with a fit index value greater than the 95th percentile of the RMSEA distribution – assuming the user's model is correct – would be viewed as misspecified.

FCOs differ from the logic in HB because only the distribution of fit indices when the model is assumed to be correct is considered whereas HB compares distributions of fit indices assuming the model is correct and assuming the model misspecified (DFI similarly compares two distributions like HB). That is, FCOs are designed to detect whether the fit of the user's model is different from a *correct* model rather than quantifying the magnitude of misspecification. FCO is essentially a parametric version of a Bollen-Stine bootstrap (Bollen & Stine, 1992) for exact fit that changes the target quantity from the chi-square statistic to a fit index. There is no inherent issue with this approach because there are merits to assessing exact fit hypotheses with a null distribution tailored to the model characteristics, but HB cutoffs will not be recovered if the FCO method is applied to model used in HB. FCO cutoffs will be stricter

because the definition for tolerable misfit is stricter – FCO cutoffs are intended to be sensitive to any deviation from exact fit whereas HB is intended to be sensitive only to a misspecification large enough to warrant concern but not to misspecifications considered acceptably small. The FCO R package facilitates implementation of the FCO method (Niemand & Mai, 2023).

DFI Strengths and Limitations

Simulations have shown that DFI decidedly outperforms HB cutoffs when characteristics do not match those studied by HB. For instance, McNeish and Wolf (2022) found that when applying one-factor models to data generated from a two-factor model, DFI cutoffs correctly classified the model as misspecified 96% of the time ($SD = 3.1\%$) across SRMR, RMSEA, and CFI with correct classification rates never falling below 90% in any individual condition. Conversely, average rejection rates for HB cutoffs were 0% for SRMR ($SD = 0.3\%$), 69% for RMSEA ($SD = 3.8\%$), and 42% for CFI ($SD = 39.2\%$). McNeish (2023) found similar performance in the context of the multifactor models – DFI cutoffs for RMSEA correctly classified 98% ($SD = 1.8\%$) of misspecified models, on average, whereas the HB cutoffs for RMSEA correctly classified 69% ($SD = 40.73\%$), on average. Importantly, note the differences in the standard deviation of the classification rates between DFI and HB cutoffs – DFI is able to consistently maintain strong performance across various conditions because it actively adapts to each model's conditions whereas performance of HB cutoffs is highly variable due to the confounding of model characteristics with misfit.

The main drawback of the DFI approach is that the algorithm used by DFI to identify hypothetical misspecifications must be redesigned for each type of model to which it is applied. Methods discussed in the previous section (FCO, simsem, and ROC-based simulation) avoid this issue because they require users to identify the hypothetically misspecified models or avoid hypothetically misspecified models altogether. However, because DFI attempts to identify hypothetical misspecifications internally to remove this burden from empirical researchers, it is less generalizable because different hypothetical misspecifications are appropriate for different

contexts. For instance, the DFI algorithm for multifactor models mimics HB and tries to identify potentially omitted cross-loadings. Conversely, the DFI algorithm for one-factor models considers potential residual correlations because one-factor models cannot have cross-loadings given that there is no other factor upon which items could cross-load.

Currently, DFI has been conceived for CFA models but there have been no proposed extensions for hierarchical factor models. Further, McNeish and Wolf (2021) note that the DFI algorithm for multifactor models may be inappropriate for hierarchical factor models because it would not differentiate between measurement misspecifications (i.e., paths from first-order factors to items) from structural misspecifications (i.e., paths from second-order factors to first-order factors). The next section proposes an extension of DFI for hierarchical factor models to broaden the scope of models supported by DFI.

Extending DFI to Hierarchical Factor Models

A primary challenge extending DFI to hierarchical factor models is that there are more types of hypothetical misspecifications to consider. That is, whereas CFA models only need to consider measurement misspecification, hierarchical models must consider misspecification between second-order factors and first-order factors (structural misspecification) or between first-order factors and observed variables (measurement misspecification). There are also more restrictions on the types of misspecifications that would make theoretical sense. That is, though sometimes permissible theoretically, researchers might not wish to generally consider that a second-order factor could load directly on an observed item without an accompanying theory, so DFI for hierarchical factor models would need to place some restrictions on the types of paths to consider adding to create general hypothetical misspecifications (the user's model is permitted to include paths from a second-order factor directly to an observed item). Two separate algorithms are therefore needed to accommodate these nuances: one for measurement misspecification and one for structural misspecification.

Measurement Misspecification

Measurement misspecifications affect relations between the first-order factors and the observed items. This situation locally represents a standard multifactor CFA model such that observed items load onto latent factors and hierarchical factor models must have multiple first-order factors. Therefore, much of the work on DFI for multifactor CFA can be applied to measurement misspecification in hierarchical factor models. In multifactor CFA, to mimic HB as closely as possible, DFI identifies the observed item with the weakest loading. The misspecified data generation model in the DFI simulation will contain a cross-loading to this item originating from the first-order factor with the highest construct reliability upon which the item did not originally load. Construct reliability is not essential to defining the misspecification, it is just used to facilitate reproducibility and systematicity as opposed to alternatives like randomly selecting a factor.

The magnitude of the standardized cross-loading for this item is equal to

$$\min \left[\lambda_o, 0.95 \times \left(\sqrt{\lambda_o^2 \psi_{(F_o, F_c)}^2 + \theta_o} - \lambda_o \psi_{(F_o, F_c)} \right) \right] \quad (1)$$

where λ_o is the estimated standardized loading of the weakest loading item in the user's model, θ_o is the estimated standardized residual variance of the weakest loading item in the user's model, and $\psi_{(F_o, F_c)}$ is the correlation between the first-order factor on which the weakest loading item originally loaded (F_o) and the first-order factor from which the cross loading will originate (F_c). Unlike multifactor CFA, $\psi_{(F_o, F_c)}$ is not directly estimated in the model but instead is implied by loadings from the second-order factors to the first-order factors. For the time-being, it is assumed that the second-order factor structure is correct. Unlike DFI for multifactor CFA, factors considered for F_c must be first-order factors; second-order factors are not considered.

Essentially, the formula will set the magnitude of the cross-loading to the same value as the loading of the weakest item if it is possible to do so without creating a non-positive definite population covariance matrix in the DFI simulation. If that is not possible, the magnitude of the

cross-loading is set to the second term instead, which is 95% of the maximum allowable cross-loading that could be added to the weakest item in the user's model without creating a non-positive definite population covariance matrix. This term uses 95% to allow some wiggle room for sampling variability (additional details on these calculations are provided in McNeish & Wolf, 2021). Multiple omitted cross-loadings can be considered as well, and the default number considered by the current software is the number of first-order factors minus 1.

In summary, DFI for measurement misspecification looks a lot like DFI for multifactor CFA except that (a) second-order factors are excluded from being the origin of cross-loadings and (b) first-order factor correlations are implied by second-order loadings in the user's model rather than being directly estimated as a parameter as in CFA. This function is included in the `dynamic` R package for deriving DFI cutoffs and is named `hier1HB`.²

Structural Misspecification

Structural misspecifications affect relations from second-order factors to first-order factors. Unlike first-order factors that are the focus of the measurement misspecification context, there may only be one second-order factor (e.g., where there is a general second-order factor). Therefore, creating hypothetical misspecifications with potentially omitted cross-loadings is not sufficiently general because such an approach would require at least two second-order factors. Instead, creating hypothetical misspecifications with disturbance correlations (i.e., residual correlations for factors) is the most general because it can be applied when there are one or more second-order factors. This is related (but not identical) to how DFI creates hypothetical misspecifications for one-factor CFA models where the same problem exists.

The magnitude of the disturbance correlations added to create hypothetical misspecifications is 0.50. Similar to DFI for one-factor CFA, the number of disturbance correlations added to the model will be a function of the number of first-order factors. Three

² Beta versions of functions described here can be downloaded from the `dynamic` GitHub package, <https://github.com/melissagwolf/dynamic> prior to being pushed to the CRAN version of the `dynamic`, which will occur after this article is published.

increasingly severe levels of misspecification will be tested, so long as the number of first-order factors is large enough to accommodate it. Level-1 includes hypothetical disturbance correlations between approximately 1/3 of first-order factor pairs, Level-2 includes hypothetical disturbance correlations between approximately 2/3 of first-order factor pairs, and Level-3 includes hypothetical disturbance correlations between all first-order factor pairs.

Mathematically, the exact number of disturbance correlations is equal to $\left\lfloor \left\lfloor F^{(1)} / 2 \right\rfloor / 3 + 0.50 \right\rfloor$ for Level-1, $\left\lfloor 2 \left\lfloor F^{(1)} / 2 \right\rfloor / 3 \right\rfloor$ for Level-2, and $\left\lfloor F^{(1)} / 2 \right\rfloor$ for Level-3 where $\lfloor \cdot \rfloor$ is the floor function that rounds the argument down to the nearest integer and $F^{(1)}$ is the number of first-order factors. At least 6 first-order factors are needed to consider all three levels (i.e., Level-1 would consider 1 disturbance correlation, Level-2 would consider 2 disturbance correlations, and Level-3 would consider 3 disturbance correlations). Models with 4 or 5 first-order factors can test Level-1 and Level-2 misspecifications. Models with 2 or 3 first-order factors cannot be tested for structural specifications because the first-order covariance structure is not overidentified. Strictly for reproducibility, first-order factors are paired together based on the magnitude of their second-order loadings from smallest to largest. This function is included in the `dynamic` R package for deriving DFI cutoffs and is named `hier2`.

Revisiting Empirical Example

Returning to the hierarchical factor model for the WISC-V from Reynolds and Keith (2017), we can reevaluate the fit of the model using the proposed DFI approach for hierarchical factor models discussed in the previous section. As a reminder, the fit indices for this model were $RMSEA = 0.033$ and $CFI = 0.985$, which satisfies HB cutoffs. However, these cutoffs may be questionably applicable to this model that features extensive differences from the models used to derive the HB cutoffs. We therefore apply DFI in an attempt to derive cutoffs that are more appropriate for this hierarchical factor model by using customized simulations tailored to the specific model characteristics. The DFI cutoffs from our proposed hierarchical DFI extension

are provided in Table 1, which are discussed in subsequent subsections. R code used to produce these results is included on the Open Science Framework page associated with this project, <https://osf.io/m4c6z/>.

DFI Cutoffs for Measurement Misspecification

DFI cutoffs designed to detect misspecifications in the measurement model are shown in the top half of Table 1. For a model with five first-order factors, hierarchical DFI will provide cutoffs for four different levels of increasingly severe misspecification based on hypothetically omitted cross-loadings. Whereas HB provides a single cutoff for each index, DFI provides multiple cutoffs per index to give researchers a range of options (called “levels” in DFI) so that fit can be evaluated more holistically. Fit indices were originally intended to operate more like effect sizes rather than as ad hoc hypothesis tests with a binary conclusion of “good” or “bad” fit, so cutoffs associated with different DFI levels are intended to provide more of a continuum for fit assessment when possible.

The magnitude of cross-loadings used as a hypothetical misspecification in the DFI simulation are reported in the rightmost column of Table 1. HB’s magnitude of misspecification roughly corresponds with the Level-1 DFI cutoffs in the first row of Table 1, which are designed to be sensitive to an omitted 0.41 standardized cross-loading. Cutoffs for Level-2 and above are not intended to map onto any terminology from HB, but each level can roughly be thought of as multiplying the misspecification magnitude in HB by the level number (e.g., the Level-3 cutoffs are intended to be sensitive roughly 3 times as large as the misspecification studied by HB).

Under the characteristics of this specific hierarchical factor model, the Level-1 DFI cutoffs that are sensitive to omitting one first-order 0.41 cross-loading are $RMSEA \leq 0.035$ and $CFI \geq 0.984$ (SRMR is also provided in Table 1, but it was not reported by Reynolds and Keith (2017), so we do not discuss it here). Notably, the DFI cutoffs have become stricter than HB cutoffs to reflect how characteristics of this model differ from those in HB’s models (e.g., slightly weaker loadings, more factors, fewer items per factor). The fitted model’s RMSEA is below the

Level-1 DFI cutoff and the fitted model's CFI is above the Level-1 cutoff, so the ultimate conclusion did not change – the cumulative misfit in this model is less than the misspecification HB considered to be meaningful. However, the margin by which the fitted model surpasses the DFI cutoffs is much slimmer than the margin by which the fitted model surpassed the HB cutoffs. It is not difficult to see how conclusions could easily change once cutoffs are recalibrated to align with the characteristics of the model being evaluated.

The Level-1 DFI cutoffs are intended to essentially reproduce the magnitude of misspecification in HB's study, but it is possible that a larger misspecification may also be considered acceptable. For instance, in a larger model with 30 items, perhaps a single omitted cross-loading is not as impactful as when there are only 10 items. Therefore, the second row of Table 1 shows the Level-2 DFI cutoffs, which are designed to be sensitive to two hypothetically omitted cross-loadings whose magnitudes are equal to 0.49 and 0.41 (i.e., the Level-1 misspecification plus a second cross-loading determined by re-running the DFI algorithm, accounting for the Level-1 misspecification). The magnitudes of the cross-loadings are different because the Level-2 cross-loading involves a different item, a different factor, and a different factor correlations, all of which influence the value of the cross-loading that roughly corresponds to what was used by HB.

The Level-2 cutoffs are more lenient at $RMSEA \leq 0.045$ and $CFI \geq 0.979$ because they are designed to be sensitive to a larger misspecification. Similarly, the Level-3 DFI cutoffs in the third row of Table 1 are designed to be sensitive to a misspecification consisting of 3 omitted cross-loadings (a 0.41 cross-loading plus the two cross-loadings from the Level-2 misspecification) and again are more lenient, as are the Level-4 DFI cutoffs which are designed to be sensitive to 4 omitted cross-loadings.

To be clear, these cross-loadings are hypothetical misspecifications used to understand the sensitivity of fit indices to misspecifications under these model characteristics. If the DFI cutoffs were exceeded, it does not imply that the model necessarily has these paths omitted or

that fit would improve if these paths were added to the model. Instead, it would just mean that the cumulative misspecification throughout the model exceeds the amount consistent with omitting these cross-loadings. The true misfit in the model could be attributable to different sources (e.g., residual correlation, incorrect factor structure) but – whatever the misspecification – the cumulative magnitude as measured by SRMR, RMSEA and CFI would exceed the hypothetical misspecifications used to derive the DFI cutoffs.

Together, these different levels are intended to give researchers some discretion when evaluating fit so that researchers can determine which magnitude of misspecification they might define as “acceptable”. There is not a single DFI level that should be uniformly consulted as the definitive cutoff in all situations and different DFI levels may be considered appropriate depending on different contexts. This is intended to help address some of the criticisms in Marsh et al. (2004) pertaining to the fact that different misspecification magnitudes may be acceptable in different circumstances (p. 330). In the previously mentioned context of a model with 30 items, although the Level-3 DFI cutoffs encode a misspecification much larger than what was studied by HB, the proportionally larger Level-3 misspecification might represent a more appropriate threshold for “acceptable” misfit for a larger model than the Level-1 cutoff. Consequently, satisfying the Level-3 DFI cutoffs may be viewed as presenting evidence of reasonable fit.

Cutoffs associated with different DFI levels are also intended to help researchers use fit indices more continuously rather than modifying models to limbo beneath a single cutoff. For instance, fit index values above the HB cutoffs are often interpreted as indicating poor fit and may encourage ad hoc model modifications to slither fit indices under the cutoffs. Conversely, a fit index value above the Level-1 DFI cutoff (roughly the same misspecification magnitude as the HB cutoff) may still be viewed as having adequate fit if the fit indices are below the Level-2 cutoffs if the hypothetical misspecifications used in the Level-2 simulation are within the researcher’s tolerance for misfit (depending on the context and model characteristics). The

presence of additional cutoffs is intended to permit more holistic evaluations of the researcher's intended model rather than lumping all fit indices above a single cutoff as equally bad and tacitly encouraging ad hoc model modification for the purpose of preserving the appearance of good fit. Broadly speaking, DFI cutoffs are an attempt to move researchers away from strict adherence to fit index cutoffs and towards using fit indices as one piece of evidence by which broader validity arguments can be built.

Figure 3 shows the visual representation of how DFI cutoffs for RMSEA work in this example. The dark grey distribution in each panel is the distribution of RMSEA from the DFI simulation when data are generated exactly from the fitted model. The light grey distribution in each panel shows the distribution of RMSEA from the DFI simulation when the fitted model is applied to data generated from the original fitted model plus the cross-loadings shown in the rightmost column of Table 1. The dotted line is the RMSEA from the empirical data. The thick dashed line shows the RMSEA value that would correctly classify 95% of models that hypothetically misspecified while incorrectly classifying no more than 5% of models that are hypothetically correct.

Again, the logic is the same as HB; however, these distributions are being uniquely tailored to characteristics in Reynolds and Keith's model. The light grey distributions (and the associated DFI cutoff) shift to the right as the DFI level increases because the misspecification is increasingly severe. This helps evaluate the empirical model by contextualizing the magnitude of the empirical RMSEA against the RMSEA associated with hypothetical misspecifications under the exact same model characteristics rather than RMSEA values associated with HB's hypothetical misspecification in HB's model characteristics.

Specifically, Figure 3 shows that the empirical RMSEA is a too large for the model to be considered correct (the dotted line is beyond the extreme upper tail of the dark grey distribution); nonetheless, the empirical RMSEA is below the values typically seen if a cross-loading were omitted (the dotted line is below the extreme lower tail of the light grey

distribution in the upper left panel). Therefore – if the hypothetical misspecification used in the Level-1 DFI simulation is considered to represent a misspecification magnitude where a researcher would begin to be concerned about misfit – this model would show reasonable approximate fit because the empirical RMSEA value is below typical RMSEA values associated with the misspecification at which the researcher would begin to be concerned. To reiterate, defining a “misspecification magnitude where a researcher would begin to be concerned” can be a challenge for empirical researchers with complex multivariate models. Whereas other methods involve researchers defining this themselves, DFI attempts to identify such a misspecification internally based on a mix of HB’s definition and the characteristics of the user’s model.

DFI Cutoffs for Structural Misspecification

DFI cutoffs for structural misspecification are shown at the bottom of Table 1. HB’s simulation on CFA did not include the types of structural misspecification that can occur in hierarchical factor models, so Level-1 cutoffs for structural misspecification in Table 1 do not correspond to HB as directly as in the measurement misspecification portion of Table 1.

For a model with 5 first-order factors, two levels of structural misspecification can be assessed. The Level-1 DFI cutoffs are sensitive to an omitted 0.50 first-order disturbance correlation. In this model, the Level-1 cutoff for structural misspecification corresponds to $RMSEA \leq 0.023$ and $CFI \geq 0.992$. The fitted model’s fit indices do not meet these cutoffs, so the conclusion would be that the cumulative misspecification in the model is greater than an omitted 0.50 disturbance correlation between first-order factors.

The Level-2 DFI cutoffs are designed to be sensitive to omitted 0.50 disturbance correlations between two pairs of first-order factors, which for these model characteristics would be $RMSEA \leq 0.033$ and $CFI \geq 0.986$. The fitted model’s RMSEA and CFI are essentially equal to the Level-2 cutoffs, so the conclusion would be that the cumulative misspecification in the model is about equal to two omitted 0.50 disturbance correlations. Again, the DFI cutoffs are not suggesting that this misspecification is present, these misspecifications are just being

used to better understand the sensitivity of fit indices under these conditions. Similar to the measurement misspecification section, providing a set of cutoffs that are sensitive to different magnitudes of misspecification is designed to encourage researchers to move away from treating fit indices as binary arbiters of fit and instead use them more continuously, as they were originally intended.

Conclusions for Hierarchical Fit Evaluation with DFI

This example shows that fit indices scale differently in hierarchical factor models than in the CFA models considered by HB. Though the conclusions may not change in this example because the fitted model surpassed the Level-1 measurement DFI cutoffs by a thin margin, this example shows how traditional HB cutoffs do not necessarily apply to hierarchical factor models and that HB cutoffs are not necessarily sensitive to misspecifications in these models, which warrants cutoffs that are specifically dedicated to hierarchical factor models.

Nonetheless, we do not know the “true” model in this empirical example, so it is impossible to determine whether the proposed DFI cutoffs for hierarchical factor models are actually more accurate than HB cutoffs because we do not know whether the model truly fits or not. To address this question directly and systematically, we conduct a simulation in the next section.

Simulation Study Design

To systematically assess the ability of DFI cutoffs to correctly classify well-fitting and misspecified models, we conduct a simulation where we can know the true status of a model’s suitability for the data. We employ two separate simulations to separately assess properties of DFI cutoffs for detecting measurement misspecifications and structural misspecifications. Although the conditions and data generation models are different for each simulation study, the general idea of each simulation study is the same.

We will generate many datasets from a particular model. Then, to each generated dataset, we fit (a) the correct model used to generate the data or (b) a model that we consider to

be meaningfully misspecified. For each fitted model, we will record the values of SRMR, RMSEA, and CFI, which are compared to both HB cutoffs and Level-1 DFI cutoffs. Level-1 DFI cutoffs are used because they are intended to be similar to HB cutoffs (at least for measurement misspecification), so the performance of each cutoff should be roughly comparable if cutoffs are functioning correctly (i.e., performance differences are not due to each method defining “acceptable” misspecification vastly differently). This also mirrors how HB and DFI have been compared in previous simulations (e.g., McNeish & Wolf, 2022).

The main outcome is whether HB and DFI cutoffs correctly classify models that are consistent with the data as well-fitting and correctly classify models that are inconsistent with the data as fitting poorly. It is also of interest whether these cutoffs perform consistently across different conditions such that identification of misspecification is not confounded with model characteristics. Details of each simulation study are described in the subsections that follow.³

Although there would be benefits to including additional methods that we previously discussed in the simulation, we did not include other methods because of potential difficulties with comparing them to HB cutoffs. For instance, the definition of tolerable misfit with the FCO method is quite different from HB cutoffs (i.e., exact vs. approximate fit), which would make comparison of FCOs to HB cutoffs less straightforward because the two methods have different definitions of what constitutes misfit and are intended to identify different things (DFI and HB have similar definitions of misfit, so they are more easily comparable). This could confound comparisons because differences in performance could be attributable to different sources (e.g., definition of misfit is different vs. ability to generalize across model characteristics).

Likewise, the ROC-based simulation method requires the user to provide a hypothetical misspecification to test, which would complicate comparisons or formation of general conclusions about performance of the ROC-based method given that results would heavily

³ Data files with simulation results for each condition are available from the first authors Open Science Framework page for this project, https://osf.io/m4c6z/?view_only=0c9f8571054c40bc8bbfa5f3ef93b17f

depend on which misspecification were selected by the researcher. Additionally, it would be unclear if readers could expect that same level of performance if they implemented the method themselves given the researcher's ability to select a meaningful misspecification to test would seem to factor into performance, so comparisons of the ROC-based method with HB or DFI may be more productive once more guidance has been developed for how users should define misspecifications with the ROC-based method.

Simulation 1: Measurement Misspecification

The measurement misspecification simulation is designed to evaluate whether HB and DFI can detect misspecification between first-order factors and observed items. We evaluate performance under two sets of conditions – one set that is methodologically oriented and another set that is practically oriented.

Methodologically Oriented Conditions

In the methodologically oriented conditions, loadings are constant across all items to identify any systematic trends in the performance of fit indices across conditions. We generate data from a model with one second-order factor and four first-order factors. These were chosen because many hierarchical models test a single general factor (like the empirical example) while Jackson et al. (2009) reported that reviewed psychology studies typically have between 3 and 6 first-order factors, so 4 falls in the middle of this range. In the first simulation design, the loadings from the second-order factor to first-order factors were all 0.75, which implies first-order factor correlations of 0.56. The loadings from the first-order factors to items have two conditions – 0.60 or 0.80 – and were constant for all items. The number of items was either 16 (4-items per first-order factor) and 32 (8-items per first-order factor), which were chosen to roughly correspond to review studies that find that psychological scales typically have between 5 and 7 items per construct (e.g., D'Urso et al., 2022; Flake et al., 2017). Sample sizes of 500 and 1000 were used in both conditions, which roughly correspond to the median and 85th percentile of the sample size distribution of CFA models in the review by Jackson et al. (2009). We chose

values towards the upper end of the distribution because hierarchical factor models tend to be larger than a typical CFA and are typically supported by larger samples. We conducted 200 replications in each combination of conditions. The number of replications is somewhat small to keep computational times manageable because DFI is a simulation-based method and each replication contains several thousand sub-replications. The computational overhead is also why we do not manipulate the second-order loadings or the number of factors in this set of conditions.

The misspecification we consider is omitted cross-loadings between first-order factors and observed items. In HB, the misspecification was a single omitted 0.50 standardized cross-loading. We deviated from this for two reasons. First, our interest in hierarchical factor models leads to some models with more items such that a single omitted cross-loading may no longer represent a meaningful misspecification (i.e., the model size problem; Shi et al., 2018). Second, we manipulate the factor loading magnitude whereas HB kept their factor loading magnitudes fixed. In our 0.80 standardized loading condition, it is not possible to add a cross-loading with a standardized magnitude near 0.50 without creating a nonpositive definite covariance matrix.

Instead, we considered using several smaller cross-loadings such that the cumulative misspecification was sufficiently large and able to be implemented across all of our conditions. To choose values, we recreated HB's simulation but changed the number and magnitude of omitted cross-loadings to find a combination of specifications that would have produced the same cutoffs that HB found in their conditions. We found that standardized cross-loadings of 0.25 on approximately 25% of items (4 out of 15) led to the same 0.06 cutoff for RMSEA and 0.95 cutoff for CFI in HB's conditions, suggesting that this misspecification is roughly consistent with a single 0.50 standardized cross-loading used by HB.

Therefore, in the 16-item condition, the misspecified data generation model includes four 0.25 cross-loadings not included in the fitted model. The 32-item condition will have 8 such cross-loadings. The fitted model is then inconsistent with the generated data and does not

include any of these cross-loadings. This misspecification results in fit indices that were not obviously indicative of good or bad fit such that consulting a cutoff for guidance would be reasonable – SRMR was between 0.03 and 0.07, RMSEA was between 0.02 and 0.10, and CFI was between 0.92 and 0.99. In the condition where the fitted model is correct, the data generation model will not feature any cross-loadings (i.e., the fitted model is the same across all conditions, but its accuracy depends on the data generation model).

Practically Oriented Conditions

In the practically oriented conditions, we generate data from a wide range of first-order and second-order loadings within the same model. This makes it more difficult to isolate particular patterns related to factor loading magnitude, but it gives a more realistic impression of how cutoffs perform when a model contains a variety of different loading magnitudes. This is similar to the empirical example where the second-order loadings range between 0.51 and 0.97 and the first-order loadings range between 0.41 and 0.85.

In this set of conditions, we kept the number of items per factor constant at 5 but manipulated the number of first-order factors to be 4, 5, or 6 such that the total number of items was 20, 25, or 30. We continued to use a single second-order factor. In the 4-factor condition, the second-order loadings were [0.45, 0.60, 0.75, 0.90], in the 5-factor condition [0.50, 0.60, 0.70, 0.80, 0.90], and in the 6-factor condition [0.50, 0.60, 0.70, 0.75, 0.80, 0.90]. The first-order loadings were [0.50, 0.60, 0.70, 0.80, 0.90] per factor in all conditions. The sample size conditions were smaller at 300 and 500 in this design to better understand where DFI might begin to breakdown and fail to provide cutoffs due to excessive sampling variability.

The number of replications was again 200 per condition. Misspecified models were created in a similar fashion with a small change. Rather than include 0.25 standardized cross-loadings on 25% of items, we included 0.375 standardized cross-loadings on 20% of items. This change was made to better generalize across the conditions for the number of items and number of first-order factors in this design. As in the methodologically oriented conditions, this

misspecification resulted in fit index values that were not obviously indicative of good or bad fit such that guidance from a cutoff would be reasonable – SRMR was between 0.05 and 0.08, RMSEA was between 0.04 and 0.07, and CFI was between 0.92 and 0.97. Ranges were smaller in these conditions than in the methodically oriented conditions because the mix of factor loading magnitudes led to less extreme swings across conditions.

Simulation Outcomes

The methodologically and practically oriented conditions both have three primary simulation outcomes that will be reported and discussed,

1. The proportion of replications that incorrectly reject a model that is consistent with the data generation model (similar to evaluating the Type-I error rate). This will assess whether DFI and HB cutoffs merely reject any model to which they are applied. Fit index cutoffs are not inferential statistics, so there is no expected value for this outcome (e.g., the false positive is not expected to be 5% as with a true Type-I error rate). Nonetheless, percentages closer to 0% are desirable on this metric.
2. The proportion of replications that correctly reject a model that is inconsistent with the data generation model (similar to evaluating power). This will assess whether DFI and HB cutoffs are sensitive to misspecifications in specific combinations of conditions. Percentages closer to 100% are desirable on this metric.
3. The proportion of replications for which DFI cutoffs can be computed with at least 90% sensitivity to hypothetical misspecification. One disadvantage of custom cutoff derivations for approximate fit is that fit index distributions of correct and misspecified models may overlap too much such that it is not possible for a single value to reliably distinguish fit indices from acceptable and misspecified models. McNeish and Wolf (2022) noted how this can occur in CFA with small samples, weak loadings, or few items. We assess the proportion of replications for which DFI cutoffs can be computed to evaluate the dependability of deriving DFI cutoffs for hierarchical factor models. This outcome is not applicable to fixed HB cutoffs.

Simulation 1 Results

Correct Models

Table 2 reports results from conditions where the fitted model was consistent with the data generation model and shows the percentage of replications where DFI cutoffs incorrectly rejected a correct model. Cells with no parenthetical values indicate that DFI cutoffs could be calculated for all 200 replications; a parenthetical value indicates the percentage of replications for which DFI cutoffs could be calculated (e.g., a cell with 91 in parentheses indicates that DFI

cutoffs were available for 91% of replications and were unavailable for 9%). HB cutoffs are not included in Table 2 because they rejected 0% of correct model replications across all conditions.

Table 2 shows that DFI cutoffs for hierarchical factor models rarely misclassified correct models. Misclassification rates were 0% in many conditions and did not exceed 4% in any single condition. Overall, this indicates the DFI cutoffs for hierarchical factor models are not blindly rejecting any model they encounter. Table 2 also corroborates findings from McNeish and Wolf (2022) regarding sampling variability precluding the ability to derive DFI cutoffs. There were few or no issues with DFI cutoff availability when $N = 500$ or $N = 1000$, especially with uniformly strong loadings. However, availability of DFI cutoffs became more problematic with lower sample sizes or weaker loadings. Expectedly, Table 2 also indicates that larger models with smaller samples also may increase sampling variability to a point where DFI cutoffs cannot be calculated (i.e., the 6-factor condition with $N = 300$ had the highest DFI cutoff unavailability).

Misspecified Models

Table 3 shows the percentage of replications in which DFI and HB cutoffs correctly classified a misspecified model as poorly fitting. Similar to Table 2, parenthetical values indicate the percentage of replications in which a DFI cutoff was available.

Regarding HB cutoffs, as has been observed in CFA models, performance of HB cutoffs for RMSEA and CFI is strongly associated with model characteristics. For instance, in the top half of Table 3, when loadings were consistently 0.60, HB cutoffs for RMSEA correctly classified 0% of misspecified models. However, when loadings were consistently 0.80, RMSEA correctly classified 100% of models. In other words, HB cutoffs confound model characteristics with misfit and HB cutoffs for RMSEA encode different information depending on the model characteristics. Similarly, the bottom half of Table 3 shows model size effects on CFI. HB cutoffs for CFI range from correctly classifying 50% of replications or fewer for 4-factor models to 100% of replications for 6-factor models, even though the misspecification is the same (proportional to model size) in all conditions. A final observation about HB cutoffs is that the HB cutoff for

SRMR has essentially no ability to identify measurement misspecification in hierarchical factor models.

On the other hand, DFI cutoffs in Table 3 were consistently able to correctly identify misspecified models across conditions and did not appear to confound model characteristics with evaluation of fit. DFI cutoffs for SRMR, RMSEA, and CFI correctly classified as least 83% of misspecified replications across all conditions and the classification rates did not systematically vary based on sample size, number of items, model size, or factor loadings. In 91% of possible conditions (38 out of 42 cells), DFI's correct classification rate was 90% or higher and the classification rate was 95% or higher in 79% of conditions (33 out of 42). DFI cutoff unavailability was also less common in Table 3 than in Table 2 and DFI cutoffs were available in a majority of replications across all conditions, even with $N = 300$. Overall, DFI cutoffs vastly outperformed HB cutoffs at identifying misspecified hierarchical factor models and consistently maintained this heightened performance across conditions. This seems to indicate that the hierarchical extension of DFI for identifying measurement model misspecifications is relatively trustworthy and attains similar performance as versions of DFI developed for CFA.

Simulation 2: Structural Misspecification

A unique feature of hierarchical factor models compared to standard CFA is that there can also be misfit in the structural model where the second-order factor(s) structures the covariances between the first-order factors. In Simulation 1, we assumed that the second-order factor structure was correct and that the first-order factor structure may have been misspecified. In Simulation 2, the situation is reversed such that the first-order factor structure is assumed to be correct, but the misspecifications occur in the second-order model. Most fit index simulation studies are based exclusively on measurement models (similar to Simulation 1) and there is less focus on structural misspecification, so there is less previous literature to guide expectations about how fit indices may perform at identifying a structural misspecification or even which

misspecifications are generally considered meaningful to assess. For these reasons, Simulation 2 is less definitive and more exploratory than Simulation 1 and also includes fewer conditions.

The conditions in Simulation 2 are similar to those used in the practically oriented conditions in Simulation 1. In Simulation 2, the second-order factor loadings are more heavily emphasized, so we chose to use simulation conditions that varied the values of these paths. We generate data from a model with 4, 5, or 6 first-order factors that each have five items. The first-order loadings were [0.50, 0.60, 0.70, 0.80, 0.90] per factor and there are three sample size conditions: 300, 500, and 1000.

Similar to Simulation 1, we have two data generation models. In the first, there is one second-order factor. In the 4-factor condition, the second-order loadings were [0.45, 0.60, 0.75, 0.90], in the 5-factor condition [0.50, 0.60, 0.70, 0.80, 0.90], and in the 6-factor condition [0.50, 0.60, 0.70, 0.75, 0.80, 0.90]. In the second condition, there are two second-order factors. The second-order loadings from the first factor are the same as described in the previous sentence, but the second factor has (a) 0.50 loadings on 2 first-order factors in the 4-factor conditions, (b) 0.45 loadings on 3 first-order factors in the 5-factor conditions, and (c) 0.50 loadings on 3 first-order factors in the 6-factor condition. The fitted model for all generated data is then a hierarchical factor model with only one second-order factor, meaning that the fitted model is consistent with the data generated from the first model but inconsistent with data generated from the second model. This again sets up a scenario in which we can evaluate the performance of DFI cutoffs to classify correct and misspecified models where the misspecification is specifying too few second-order factors.

Similar to Simulation 1, the outcomes of Simulation 2 are (a) the proportion of incorrectly rejected replications when the fitted model is consistent with the generated data, (b) the proportion of correctly rejected replications when the fitted model is inconsistent with the generated data, and (c) the proportion of replications for which DFI cutoffs are available.

Simulation 2 Results

Table 4 shows the results for Simulation 2; the top half shows results when the fitted model is consistent with the data generation model and the bottom half shows results when the fitted model is inconsistent with the data generation model. Neither DFI nor HB cutoffs had any difficulty when the fitted model was correct – HB cutoffs never rejected a correct model whereas DFI cutoffs rarely rejected correct models but never more than 4% of the time. DFI cutoffs were always available for SRMR regardless of sample size, but DFI cutoffs for RMSEA and CFI were always unavailable when $N < 500$.

Regarding performance to detect misspecification, HB cutoffs performed very poorly and detected 0% of misspecified replications in all conditions. This echoes findings in Simulation 1 where HB cutoffs have little sensitivity to misspecifications in hierarchical factor models, which is not surprising given that they were not intended to generalize to these models (even though they often are anyway). Performance of DFI cutoffs for identifying structural misspecifications in Table 4 was decent, but not as strong as results for measurement misspecification in Table 3.

Under the conditions included in the simulation, DFI cutoffs for SRMR was clearly the best able to detect structural misspecification and did so in at least 86% of replications in the conditions studied. DFI cutoffs for RMSEA and CFI performed about equally and provided substantial relative improvements over HB cutoffs, but the absolute correct classification rates were somewhat modest in conditions with either smaller samples or smaller models. DFI cutoffs for RMSEA and CFI were also more likely to be unavailable than DFI cutoffs for SRMR.

The previous literature evaluating common fit indices like SRMR, RMSEA, and CFI rarely considers hierarchical factor models, but evidence from this narrow set of conditions suggests that all three indices retain some sensitivity to misspecifications between second-order and first-order factors when evaluated with DFI cutoffs (e.g., greater than 50% sensitivity in replications where DFI cutoffs are available). SRMR appears to be the most sensitive and trustworthy and the simulation suggested that it should be preferred (e.g., greater than 85%

sensitivity with minimal unavailability), at least based on the conditions we have studied. Of course, this evidence comes from a limited set of conditions and additional studies would help to further clarify the potential utility and sensitivity of these indices – whether in conjunction with DFI or not – in the context of hierarchical factor models.

These results corroborate previous studies suggesting that SRMR is more sensitive to misspecifications involving factor correlations (e.g., Fan & Sivo, 2005), which are directly affected by structural misspecification. This supports previous suggestions for the two-index strategy (Hu & Bentler, 1998), which seems especially relevant in hierarchical factor models because the factor correlations are structured and susceptible to misspecification as opposed to CFA where the typical approach is to saturate the factor covariance structure. **Simulation**

Summary

1. HB cutoffs did not incorrectly classify correct models, but HB cutoffs were largely insensitive to misspecifications in hierarchical factor models and suggested most misspecified models fit well. These findings were somewhat expected because (a) they are consistent with previous studies and (b) HB cutoffs were never intended to be generalized to hierarchical factor models (even though they often are).
2. The proposed hierarchical extension of DFI cutoffs for SRMR, RMSEA, and CFI were consistently able to detect misspecifications between first-order factors and observed items. Nearly all conditions correctly classified over 90% of replications with measurement misspecifications. There were also very few instances of DFI cutoffs incorrectly classifying incorrect models as misfitting and false positive rates were often 0% and never above 4%. The proposed DFI extensions for hierarchical factor models performed as well as previous versions of DFI developed for CFA models.
3. DFI cutoffs for structural misspecification offered large relative improvements over HB cutoffs. However, the true positive rates were lower than those seen in the conditions evaluating measurement misspecifications. These results suggest that DFI could improve current practice, but also that additional refinements could be made to augment how DFI searches for hypothetical misspecifications to simulate. It is also possible that fit indices may simply be less sensitive to structural misspecifications and have more difficulty detecting misspecification in this part of the model.
4. Similar to previous studies on DFI, we found that DFI cutoffs were sometimes unavailable. Predictably, this occurred in conditions where the sampling variability of simulated fit index distributions used to derive DFI cutoffs increased. This includes smaller samples, weaker loadings, and larger models; especially when multiple of these elements are present simultaneously. Sample sizes of 300 or 500 were able to return DFI cutoffs in a large percentage of replications in the conditions we studied, but users should note that DFI is not guaranteed to produce cutoffs that will be able to achieve the

same sensitivity to misspecification as HB cutoffs.

Discussion, Limitations, & Extensions

Jackson et al.'s 2009 review noted that 11% of factor models in psychology papers are hierarchical models with second-order factors; however, despite a sizeable volume of literature on the fit of factor models, attention has scarcely been paid to issues related to hierarchical factor models. Instead, it is common to generalize findings for CFA models to the hierarchical factor model case. In this paper, we showed that applying traditional fit index cutoffs derived for CFA like $SRMR \leq 0.08$, $RMSEA \leq 0.06$, and $CFI \geq 0.95$ to hierarchical factor models leads to poor performance and a considerable inability to identify non-trivial misspecifications. Additionally, these traditional benchmarks appeared to be confounded with model characteristics such that performance was inconsistent across various model conditions.

To address this situation, we proposed an extension of the DFI framework for hierarchical factor models. DFI expands upon previous proposals for using customized simulations to derive tailored, model-specific fit index cutoffs with the goal to make the approach more accessible to a broader audience. We conducted simulation studies to assess the performance of our proposed DFI approach for hierarchical factor models, which showed that the method (a) was consistent across various model characteristic conditions, (b) outperformed traditional cutoffs, often by large margins, (c) was highly effective for identifying measurement misspecification, and (d) was moderately effective for identifying structural misspecification.

One disadvantage is that customized cutoffs are not always available if sampling variability is high because a single fit index value that reliably differentiates between acceptable and misspecified models may not necessarily exist. To return to the power analysis analogy mentioned earlier, unavailability of cutoffs is similar to trying to form conclusions from a non-significant treatment effect when power is low. In the treatment effect context, a non-significant effect could occur because the treatment effect is null in the population but it could also occur if the effect is non-null but there is insufficient power to detect the effect. The interpretation is

similar in the DFI context—when cutoffs are unavailable, it means that it is not possible to distinguish fit indices from correct and misspecified models within a sufficient degree of certainty.

In such cases, one option may be to reduce the required sensitivity to misspecification in the DFI simulations. To retain comparability to HB cutoffs, we required 90% hypothetical sensitivity for DFI to produce a cutoff. In terms of Figure 1, this means that at least 90% of the light grey distribution would be to the right of the cutoff (note that hypothetical sensitivity is based on the internal DFI simulations is distinct from the empirical sensitivity in the simulation study, which could be below 90% as in Table 3 and 4). However, this could be reduced (e.g., 80%) if a researcher has a smaller sample or weaker loadings and anticipates availability of DFI cutoffs being problematic. Another option may be to use the ROC-based approach of Groskurth et al. (2022), which may naturally be more accommodating in conditions where DFI cutoffs are unavailable because the ROC-based method determines the optimal sensitivity and specificity without requiring users to determine a specific sensitivity with which they are comfortable.

Another interpretation of unavailable DFI cutoffs may be that global approximate fit indices may not be an ideal method to evaluate model fit because it may be unclear how to identify which values of global fit indices are indicative of “acceptable” or “reasonable” fit. If adopting this viewpoint, exact fit or examinations of local fit may be helpful. Unavailability of DFI cutoffs is attributable to high sampling variability that often occurs with smaller samples in which arguments against exact fit on the grounds of tests being overly sensitive to trivial misspecification are less persuasive.

Alternatively, evaluation of local fit by – for instance – inspecting the standardized residual correlation matrix may be useful (Appelbaum et al., 2018; McDonald & Ho, 2002, p. 73). This matrix captures the discrepancy between each element of the observed and model-implied correlation matrices and can provide information about areas of local strain (Browne et al., 2002; Kline, 2010; Millsap, 2007, p. 879). For instance, if all elements of this matrix are

close to 0, that can indicate that the model approximately fits because there are only minor deviations between the observed and model-implied correlation matrices. Conversely, if there are any large values (e.g., greater than 0.10; Schreiber, 2008, p. 91), this can suggest that there is a non-ignorable misspecification in the model. Though local fit is rarely reported in empirical studies (Ropovik, 2015; Zhang et al., 2021), it can be a valuable complement or alternative for more commonly reported global fit indices (Thoemmes et al., 2018).

As extensions and limitations, first, the simulation results for structural misspecification were decent but not great. There were a few competing reasons why this may have occurred. First, structural misspecification – especially in hierarchical factor models – is not studied nearly as often as measurement misspecification. As a result, it was less clear how to select a data generation model for this type of misspecification, and the somewhat inconsistent performance may have been due to our misspecifications not necessarily being constant across conditions. Relatedly, the misspecifications may not have been large enough to elicit a conclusion that the model was non-trivially misspecified, which may have been reflected by the fact that HB cutoffs classified 0% of replications as misspecified in all structural misspecification conditions. The structural model also contains relatively fewer degrees of freedom than the measurement model, so misfit may just be more difficult to detect (McDonald & Ho, 2002). On a similar note, there were also relatively few first-order factors in the simulation, which also limited the number and size of potential misspecifications that could be added. Overall, there was less certainty in the adequacy of the simulation design for structural misspecifications, so we cannot rule out our choices in the simulation design as one possible source of the more tempered performance when addressing structural misspecifications. Lastly, HB focused predominantly on measurement misspecification and there is less analogous information on structural misspecification upon which to base DFI. It may be possible that detection rates in the simulation were lower because the hypothetical misspecification included in the DFI algorithm was too large such that the misspecifications in the simulation did not always register as

meaningfully misspecified. In general, further understanding of structural misspecifications in hierarchical factor models would be valuable in future work and could help refine DFI algorithms to ensure that derived cutoffs correspond to practically meaningful levels of misspecification.

Second, our simulations and the empirical example assumed that item responses were continuous and that maximum likelihood estimation was applied (though do note that DFI software can accommodate other estimation methods). However, many hierarchical factor models are applied to items that are categorical such as yes/no responses or Likert responses (e.g., Flake et al., 2017; Jackson et al., 2009). Some recent work has argued that Likert-type data can defensibly be treated as continuous (Robitzsch, 2020), although others have noted that the appropriateness of doing so can depend on distributional characteristics of Likert-type item responses (e.g., Rhemtulla et al., 2012). Our simulation results are limited to treating responses as continuous and additional extensions would need to be developed to address situations where item responses are explicitly modeled as categorical (e.g., McNeish, in press). Our simulation also did not include missing data, which are common and may impact effectiveness of hierarchical DFI cutoffs.

Third, our simulation split up measurement and structural misspecification into two separate simulations. However, it is likely that models in practice will contain both types of misfit simultaneously. Additional extensions for hierarchical DFI may wish to consider a way to create hypothetical misspecifications in multiple parts of the model simultaneously. Some caution will need to be taken though, because it would be prudent to ensure that misspecification in one part of the model does not dominate or wash out misspecification in another part of the model. It would also be necessary to ensure that adding measurement and structural misspecifications simultaneously does not make the cumulative misspecification to which DFI cutoffs are sensitive excessively large.

Fourth, we mostly approached hierarchical factor model fit such that the interest was

assessing the hierarchical factor model in isolation. However, hierarchical factor models are frequently fit as one model in a sequence (usually after a correlated factors CFA) to determine if structuring the factor correlations with second-order factors fits appreciably differently than saturating the factor correlations. One option for model comparisons may be to apply DFI cutoffs to both the multifactor CFA and the hierarchical model because the measurement misspecification for DFI with hierarchical factor models is similar to the DFI multifactor CFA misspecification, so the DFI cutoffs may be comparable. Another possible way to address this is that DFI software allows users to output all simulated fit index values into a .csv file.

Researchers could apply DFI to their multifactor CFA and save the simulated fit indices, then apply hierarchical DFI to their hierarchical factor model and save the simulated fit indices. The seed value for the simulations is the same for both software applications, so the fit index differences could be subtracted to form a fit index distribution of the difference between a CFA with correlated factors and a hierarchical factor model. Additional work would be needed to assess how useful this type of approach may be, but it is one place one could start when thinking through how to integrate DFI with model comparisons. As another extension, bifactor models are also included in such sequences, so extending DFI to bifactor models to determine a suitable method to create hypothetical misspecifications would also be valuable.

Lastly, we fully recognize that the entire enterprise of determining cutoffs for fit indices is likely not the best approach for evaluating models and psychometric scales. For instance, fit indices are one way to provide internal structure validity evidence, but the standards for educational and psychological testing from AERA, APA, and NCME (2014) do not require this and list four other ways that researchers can provide validity evidence for scores that do not require fit indices. Ideally, measurement models and the scores they produce would be defended more holistically from a variety of perspectives (Kane, 2013). Nonetheless, we realize the undeniable convenience of using quantitative evidence and understand why this has become the de facto method of evaluating measurement models. Therefore, although we do not

unequivocally endorse reliance on fit indices and interpreting them based on adherence to (somewhat arbitrary) cutoffs, the last 20 years of research has shown that many researchers support this type of strategy regardless of the criticism it has received in the methodological and philosophy of science literatures. Our motivation for developing DFI further is that – if researchers insist on heavily relying upon fit index cutoffs – it is worthwhile to provide a better way to do it to improve accuracy of decisions and conclusions.

Concluding Remarks

Many empirical researchers are under the impression that the methodological literature's understanding of model fit is far more advanced than it actually is. This is not their fault since large portions of the methodological literature and methodological training empirical researchers receive emphasizes easy-to-recall cutoffs that seem universally applicable. In reality, we have an idea which values of fit indices are good for only a narrow subset of possible models. In this paper, we discussed how hierarchical factor models are not within this subset and there have been limited attempts to understand how fit indices behave in these models. We showed that generalizing traditional cutoffs to hierarchical factor models results in weak performance to correctly evaluate models. So, we proposed to adapt the framework to hierarchical factor models as one general way to obtain fit index cutoffs that can provide more accurate assessments of how well a hierarchical factor model fits. Our simulations showed that this method had promise and could provide large improvements over the status quo of generalizing cutoffs that were never meant for hierarchical factor models. Nonetheless, we hope that this paper is not the last that attempts to refine fit evaluation in hierarchical factor models because there are clear extensions and refinements that could further improve fit evaluation in these models.

References

- American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (2014). *Standards for educational and psychological testing*. American Psychological Association.
- Appelbaum, M., Cooper, H., Kline, R. B., Mayo-Wilson, E., Nezu, A. M., & Rao, S. M. (2018). Journal article reporting standards for quantitative research in psychology: The APA Publications and Communications Board task force report. *American Psychologist*, 73(1), 3-25.
- Arend, M. G., & Schäfer, T. (2019). Statistical power in two-level models: A tutorial based on Monte Carlo simulation. *Psychological Methods*, 24(1), 1–19.
- Barrett, P. (2007). Structural equation modelling: Adjudging model fit. *Personality and Individual Differences*, 42(5), 815–824.
- Bentler, P. M., & Bonett, D. G. (1980). Significance tests and goodness of fit in the analysis of covariance structures. *Psychological Bulletin*, 88(3), 588-606.
- Bollen, K. A. (1989). *Structural equations with latent variables*. Wiley.
- Bollen, K. A., & Stine, R. A. (1992). Bootstrapping goodness-of-fit measures in structural equation models. *Sociological Methods & Research*, 21(2), 205-229.
- Brown, T. A. (2015). *Confirmatory factor analysis for applied research*. Guilford.
- Browne, M. W., & Cudeck, R. (1993). Alternative ways of assessing model fit. In K. A. Bollen & J. S. Long (Eds.), *Testing structural equation models* (pp. 136-162). Sage
- Browne, M. W., & Cudeck, R. (1992). Alternative ways of assessing model fit. *Sociological Methods & Research*, 21(2), 230–258.
- Browne, M. W., MacCallum, R. C., Kim, C. T., Andersen, B. L., & Glaser, R. (2002). When fit indices and residuals are incompatible. *Psychological Methods*, 7(4), 403–421.
- Brunner, M., Nagy, G., & Wilhelm, O. (2012). A tutorial on hierarchically structured constructs. *Journal of Personality*, 80(4), 796-846.
- Carroll, J. B. (1993). *Human cognitive abilities: A survey of factor-analytical studies*. Cambridge University Press.
- Caspi, A., Houts, R. M., Belsky, D. W., Goldman-Mellor, S. J., Harrington, H., Israel, S., ... & Moffitt, T. E. (2014). The p factor: one general psychopathology factor in the structure of psychiatric disorders?. *Clinical Psychological Science*, 2(2), 119-137.
- Chen, F., Curran, P. J., Bollen, K. A., Kirby, J., & Paxton, P. (2008). An empirical evaluation of the use of fixed cutoff points in RMSEA test statistic in structural equation models. *Sociological Methods & Research*, 36(4), 462–494.
- D’Urso, D. E., Maassen, E., van Assen, M. A. L. M., Nuijten, M. B., De Roover, K., & Wicherts, J. M. (2022). The dire disregard of measurement invariance testing in psychological science.

PsyArXiv. <https://psyarxiv.com/n3f5u/>. Accessed 29 Jul 2022

DeYoung, C. G. (2006). Higher-order factors of the Big Five in a multi-informant sample. *Journal of Personality and Social Psychology*, 91(6), 1138–1151.

Digman, J. M. (1997). Higher-order factors of the Big Five. *Journal of Personality and Social Psychology*, 73(6), 1246.

Fan, X., & Sivo, S. A. (2007). Sensitivity of fit indices to model misspecification and model types. *Multivariate Behavioral Research*, 42(3), 509–529.

Fan, X., & Sivo, S. A. (2005). Sensitivity of fit indexes to misspecified structural or measurement model components: Rationale of two-index strategy revisited. *Structural Equation Modeling*, 12(3), 343–367.

Flake, J. K., Pek, J., & Hehman, E. (2017). Construct validation in social and personality research: Current practice and recommendations. *Social Psychological and Personality Science*, 8(4), 370–378.

Groskurth, Bhaktha, & Lechner (2022). Making model judgments ROC(K)solid: Tailored cutoffs for fit indices through simulation and ROC analysis in structural equation modeling. *PsyArXiv*, <https://psyarxiv.com/62j89/> (Accessed 16 May 2023).

Green, P., & MacLeod, C. J. (2016). SIMR: An R package for power analysis of generalized linear mixed models by simulation. *Methods in Ecology and Evolution*, 7(4), 493–498.

Greiff, S., & Heene, M. (2017). Why psychological assessment needs to start worrying about model fit. *European Journal of Psychological Assessment*, 33(5), 313–317.

Hancock, G. R., & Mueller, R. O. (2011). The reliability paradox in assessing structural relations within covariance structure models. *Educational and Psychological Measurement*, 71(2), 306–324.

Heene, M., Hilbert, S., Draxler, C., Ziegler, M., & Buhner, M. (2011). Masking misfit in confirmatory factor analysis by increasing unique variances: A cautionary note on the usefulness of cutoff values of fit indices. *Psychological Methods*, 16(3), 319–336.

Hu, L. T., & Bentler, P. M. (1998). Fit indices in covariance structure modeling: Sensitivity to underparameterized model misspecification. *Psychological Methods*, 3(4), 424–453.

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, 6(1), 1–55.

Hu, L. T., Bentler, P. M., & Kano, Y. (1992). Can test statistics in covariance structure analysis be trusted? *Psychological Bulletin*, 112(2), 351–362.

Jackson, D. L., Gillaspay, J. A., & Purc-Stephenson, R. (2009). Reporting practices in confirmatory factor analysis: An overview and some recommendations. *Psychological Methods*, 14(1), 6–23.

Johnson, D. R., Kaufman, J. C., Baker, B. S., Patterson, J. D., Barbot, B., Green, A. E., ... & Beaty,

R. E. (2023). Divergent semantic integration (DSI): Extracting creativity from narratives with distributional semantic modeling. *Behavior Research Methods*, advance online publication

Kane, M. (2013). The argument-based approach to validation. *School Psychology Review*, 42(4), 448–457.

Kim, H., & Millsap, R. (2014). Using the Bollen-Stine bootstrapping method for evaluating approximate fit indices. *Multivariate Behavioral Research*, 49(6), 581–596.

Kline, R. B. (2010). *Principles and practice of structural equation modeling* (3rd ed). Guilford.

Krueger, R. F., Markon, K. E., Patrick, C. J., Benning, S. D., & Kramer, M. D. (2007). Linking antisocial behavior, substance use, and personality: an integrative quantitative model of the adult externalizing spectrum. *Journal of Abnormal Psychology*, 116(4), 645–666.

Markon, K. E., Krueger, R. F., & Watson, D. (2005). Delineating the structure of normal and abnormal personality: an integrative hierarchical approach. *Journal of Personality and Social Psychology*, 88(1), 139–157.

Marsh, H. W., & Craven, R. G. (2006). Reciprocal effects of self-concept and performance from a multidimensional perspective: Beyond seductive pleasure and unidimensional perspectives. *Perspectives on Psychological Science*, 1(2), 133–163.

Marsh, H. W., & Hocevar, D. (1985). Application of confirmatory factor analysis to the study of self-concept: First-and higher order factor models and their invariance across groups. *Psychological Bulletin*, 97(3), 562–582.

Marsh, H. W., Hau, K. T., & Grayson, D. (2005). Goodness of fit evaluation in structural equation modeling. In A. Maydeu-Olivares & J. McArdle (Eds.), *Psychometrics: A festschrift to Roderick P. McDonald* (pp. 275–340). Erlbaum Associates.

Marsh, H. W., Hau, K.-T., & Wen, Z. (2004). In search of golden rules: Comment on hypothesis testing approaches to setting cutoff values for fit indexes and dangers in overgeneralizing Hu and Bentler's (1999) Findings. *Structural Equation Modeling*, 11(3), 320–341.

Martin, A. J. (2007). Examining a multidimensional model of student motivation and engagement using a construct validation approach. *British Journal of Educational Psychology*, 77(2), 413–440.

Maydeu-Olivares, A., Shi, D., & Rosseel, Y. (2018). Assessing fit in structural equation models: A Monte-Carlo evaluation of RMSEA versus SRMR confidence intervals and tests of close fit. *Structural Equation Modeling*, 25(3), 389–402.

McGrew, K. S. (2009). CHC theory and the human cognitive abilities project: Standing on the shoulders of the giants of psychometric intelligence research. *Intelligence*, 37(1), 1–10.

McDonald, R. P., & Ho, M. H. R. (2002). Principles and practice in reporting structural equation analyses. *Psychological Methods*, 7(1), 64–82.

McNeish, D. & Wolf, M.G. (2021). Dynamic fit index cutoffs for confirmatory factor analysis models. *Psychological Methods*, advance online publication

McNeish, D. & Wolf, M.G. (in press). Dynamic fit cutoffs for one-factor models. *Behavior Research Methods*, advance online publication.

McNeish, D. (2023). Generalizability of dynamic fit index, equivalence testing, and Hu & Bentler cutoffs for evaluating fit in factor analysis. *Multivariate Behavioral Research*, advance online publication.

McNeish, D. (in press). Dynamic fit index cutoffs for factor analysis with Likert, ordinal, or binary responses. <https://psyarxiv.com/tp35s/>

Meehl, P. E. (1978). Theoretical risks and tabular asterisks: Sir Karl, Sir Ronald, and the slow progress of soft psychology. *Journal of Consulting and Clinical Psychology*, 46(4), 806–834.

Miles, J., & Shevlin, M. (2007). A time and a place for incremental fit indices. *Personality and Individual Differences*, 42(5), 869–874.

Millsap, R. E. (2007). Structural equation modeling made difficult. *Personality and Individual Differences*, 42(5), 875–881.

Millsap, R. E. (2013). A simulation paradigm for evaluating model fit. In M. Edwards & R. MacCallum (Eds.), *Current issues in the theory and application of latent variable models* (pp. 165–182). Routledge.

Mulaik, S. A. (2009). *Foundations of factor analysis*. McGraw-Hill.

Murray, A. L., & Johnson, W. (2013). The limitations of model fit in comparing the bi-factor versus higher-order models of human cognitive ability structure. *Intelligence*, 41(5), 407–422.

Niemand, T., & Mai, R. (2018). Flexible cutoff values for fit indices in the evaluation of structural equation models. *Journal of the Academy of Marketing Science*, 46, 1148–1172.

Nieman, T. & Mai, R. (2023). FCO: Flexible cutoffs for model fit evaluation in covariance-based structural models. [Software]. Available from CRAN, <https://cran.r-project.org/web/packages/FCO/index.html>

Nye, C. D., & Drasgow, F. (2011). Assessing goodness of fit: Simple rules of thumb simply do not work. *Organizational Research Methods*, 14(3), 548–570.

Paxton, P., Curran, P. J., Bollen, K. A., Kirby, J., & Chen, F. (2001). Monte Carlo experiments: Design and implementation. *Structural Equation Modeling*, 8(2), 287–312.

Pornprasertmanit, S., Wu, W., & Little, T. D. (2013). Using a Monte Carlo approach for nested model comparisons in structural equation modeling. In R. E. Millsap, L. A. van der Ark, D. M. Bolt, & C. M. Woods (Eds.), *New developments in quantitative psychology* (pp. 187–197). Springer.

Pornprasertmanit, S., Miller, P., Schoemann, A., Jorgensen, T., & Quick, C. (2022). SIMulated structural equation modeling (R package version 0.5-16). <http://CRAN.R-project.org/package=simsem>

Reynolds, M. R., & Keith, T. Z. (2017). Multi-group and hierarchical confirmatory factor analysis of the Wechsler Intelligence Scale for Children—Fifth Edition: What does it measure?. *Intelligence*, 62, 31-47.

Rhemtulla, M., Brosseau-Liard, P. É., & Savalei, V. (2012). When can categorical variables be treated as continuous? A comparison of robust continuous and categorical SEM estimation methods under suboptimal conditions. *Psychological Methods*, 17(3), 354-373.

Rijmen, F. (2010). Formal relations and an empirical comparison among the bi-factor, the testlet, and a second-order multidimensional IRT model. *Journal of Educational Measurement*, 47(3), 361-372.

Robitzsch, A. (2020, October). Why ordinal variables can (almost) always be treated as continuous variables: Clarifying assumptions of robust continuous and ordinal factor analysis estimation methods. *Frontiers in Education*, 5, 589965.

Ropovik, I. (2015). A cautionary note on testing latent variable models. *Frontiers in Psychology*, 6, 1715.

Saris, W. E., Satorra, A., & Van der Veld, W. M. (2009). Testing structural equation models or detection of misspecifications?. *Structural Equation Modeling*, 16(4), 561-582.

Saris, W. E., Satorra, A., & Sörbom, D. (1987). The detection and correction of specification errors in structural equation models. *Sociological Methodology*, 17, 105-129.

Schönemann, P. H. (1981). Power as a function of communality in factor analysis. *Bulletin of the Psychonomic Society*, 17(1), 57-60.

Schreiber, J. B. (2008). Core reporting practices in structural equation modeling. *Research in Social and Administrative Pharmacy*, 4(2), 83-97.

Shi, D., Lee, T., & Terry, R. A. (2018). Revisiting the model size effect in structural equation modeling. *Structural Equation Modeling*, 25(1), 21-40.

Shi, D., Lee, T., & Maydeu-Olivares, A. (2019). Understanding the model size effect on SEM fit indices. *Educational and Psychological Measurement*, 79(2), 310-334.

Silvia, P. J. (2008). Another look at creativity and intelligence: Exploring higher-order models and probable confounds. *Personality and Individual Differences*, 44(4), 1012-1021.

Sivo, S. A., Fan, X., Witta, E. L., & Willse, J. T. (2006). The search for “optimal” cutoff properties: Fit index criteria in structural equation modeling. *Journal of Experimental Education*, 74(3), 267-288.

Tanaka, J. S. (1987). “How big is big enough?”: Sample size and goodness of fit in structural equation models with latent variables. *Child Development*, 58(1), 134-146.

Thoemmes, F., Rosseel, Y., & Textor, J. (2018). Local fit evaluation of structural equation models using graphical criteria. *Psychological Methods*, 23(1), 27-41.

Tomarken, A. J., & Waller, N. G. (2003). Potential problems with “well fitting” models. *Journal*

of Abnormal Psychology, 112(4), 578–598

West, S.G., Wu, W., McNeish, D., & Savord, A. (2023). Model fit in structural equation modeling. In R.H. Hoyle (Ed.), *Handbook of Structural Equation Modeling* (2nd Ed.) New York: Guilford Press, pp. 184-205.

Wolf, M.G. & McNeish, D. (2023). dynamic: An R package for deriving dynamic fit index cutoffs for factor analysis. *Multivariate Behavioral Research*, advance online publication.

Wolf, M.G. & McNeish, D. (2022). dynamic: DFI cutoffs for latent variables models (version 1.1.0). [Software]. Available from CRAN, <https://cran.r-project.org/web/packages/dynamic>

Wolf, M. G. & McNeish, D. (2021). Dynamic Model Fit (version 1.1.0). [Software]. Available from www.dynamicfit.app

Wright, A. G., Lukowitsky, M. R., Pincus, A. L., & Conroy, D. E. (2010). The higher order factor structure and gender invariance of the Pathological Narcissism Inventory. *Assessment*, 17(4), 467-483.

Yung, Y. F., Thissen, D., & McLeod, L. D. (1999). On the relationship between the higher-order factor model and the hierarchical factor model. *Psychometrika*, 64, 113-128.

Zhang, M. F., Dawson, J. F., & Kline, R. B. (2021). Evaluating the use of covariance-based structural equation modelling with reflective measurement in organizational and management research: A review and recommendations for best practice. *British Journal of Management*, 32(2), 257-272.

Zinbarg, R. E., Barlow, D. H., & Brown, T. A. (1997). Hierarchical structure and general factor saturation of the Anxiety Sensitivity Index: Evidence and implications. *Psychological Assessment*, 9(3), 277-284.

Zyphur, M. J., Bonner, C. V., & Tay, L. (2023). Structural Equation Modeling in Organizational Research: The State of Our Science and Some Proposals for Its Future. *Annual Review of Organizational Psychology and Organizational Behavior*, 10, 495-517.

Table 1

DFI cutoffs for measurement and structural misspecification in the hierarchical factor model in Reynolds & Keith (2017). Level-1 is intended to be sensitive to a misspecification of similar magnitude as studied by Hu and Bentler. Higher levels of DFI cutoffs are intended to be sensitive to increasingly severe misspecification. The rightmost column shows the misspecifications that are tested at each of these levels.

		SRMR	RMSEA	CFI	Hypothetical Misspecification
Measurement					
	Level-1	.032	.035	.984	0.41
	Level-2	.036	.045	.979	0.49, 0.41
	Level-3	.038	.054	.974	0.41, 0.49, 0.41
	Level-4	.039	.066	.967	0.34, 0.41, 0.49, 0.41
Structural					
	Level-1	.024	.023	.992	(Processing Speed, Working Memory)
	Level-2	.030	.033	.986	Level-1 plus (Visual, Verbal Comp.)

Table 2

Percentage of replications in which DFI cutoffs incorrectly classified a correct model as being misspecified. Values close to 0 are desirable and indicate a good ability to identify that a hierarchical model fits well. Parenthetical values indicate the percentage of replications in which DFI cutoffs were available; cells with no parenthetical values indicate that DFI cutoffs were available for all replications.

Factors	Loadings	Items	N = 300			N = 500			N = 1000		
			SRMR	RMSEA	CFI	SRMR	RMSEA	CFI	SRMR	RMSEA	CFI
4	0.60	16	---	---	---	2 (91)	2 (88)	4 (75)	0	0	0
		32	---	---	---	1	1	1	0	0	0
	0.80	16	---	---	---	0	0	0	0	0	0
		32	---	---	---	0	0	0	0	0	0
4	Mixed	20	1 (82)	1 (86)	2 (84)	2 (93)	0	2	---	---	---
5	Mixed	25	2 (65)	2 (57)	3 (55)	0 (89)	1	1	---	---	---
6	Mixed	30	3 (69)	2 (28)	0 (24)	3 (88)	0	1	---	---	---

Note: N = sample size, “---” indicates that the sample size condition was not included for the particular combination of conditions. HB cutoffs are not shown because the values were always 0 for all three indices in all conditions.

Table 3

Percentage of replications in which DFI and HB cutoffs identified that a model omitting cross-loadings from first-order factors to observed items was misspecified. Values close to 100 are desirable and indicate a strong ability to identify misspecifications in hierarchical models. Values close to 0 indicate that misspecifications were rarely correctly identified. Parenthetical values indicate the percentage of replications in which DFI cutoffs were available; cells with no parenthetical values indicate that DFI cutoffs were available for all replications.

Factors	Loadings	Items	Cutoff	N = 300			N = 500			N = 1000		
				SRMR	RMSEA	CFI	SRMR	RMSEA	CFI	SRMR	RMSEA	CFI
4	0.60	16	DFI	---	---	---	89 (58)	86(92)	93 (79)	91	83	96
			HB	---	---	---	0	0	6	0	0	0
		32	DFI	---	---	---	99	97	100	100	97	99
			HB	---	---	---	0	0	7	0	0	0
		16	DFI	---	---	---	100	98	99	100	99	100
			HB	---	---	---	0	100	24	0	100	25
4	0.80	32	DFI	---	---	---	100 (82)	100	100	100	100	100
			HB	---	---	---	0	100	100	0	100	100
		20	DFI	93 (97)	90 (97)	91 (97)	87	93	95	---	---	---
			HB	0	20	51	0	7	39	---	---	---
		25	DFI	100 (74)	96 (95)	99 (93)	100 (89)	96	98	---	---	---
			HB	4	17	87	0	3	95	---	---	---
5	Mixed	30	DFI	100 (81)	99 (94)	99 (93)	100 (89)	99	100	---	---	---
			HB	6	1	98	0	0	100	---	---	---
6	Mixed	25	DFI	100 (81)	99 (94)	99 (93)	100 (89)	99	100	---	---	---
			HB	6	1	98	0	0	100	---	---	---

Note: N = sample size, “---” indicates that the sample size condition was not included for the particular combination of conditions.

Percentage of replications in which DFI and HB cutoffs identified that a model omitted a second-order factor was misspecified. Values close to 100 are desirable and indicate a strong ability to identify misspecifications in hierarchical models. Values close to 0 indicate that misspecifications were rarely correctly identified. Parenthetical values indicate the percentage of replications in which DFI cutoffs were available; cells with no parenthetical values indicate that DFI cutoffs were available for all replications.

[illegible]

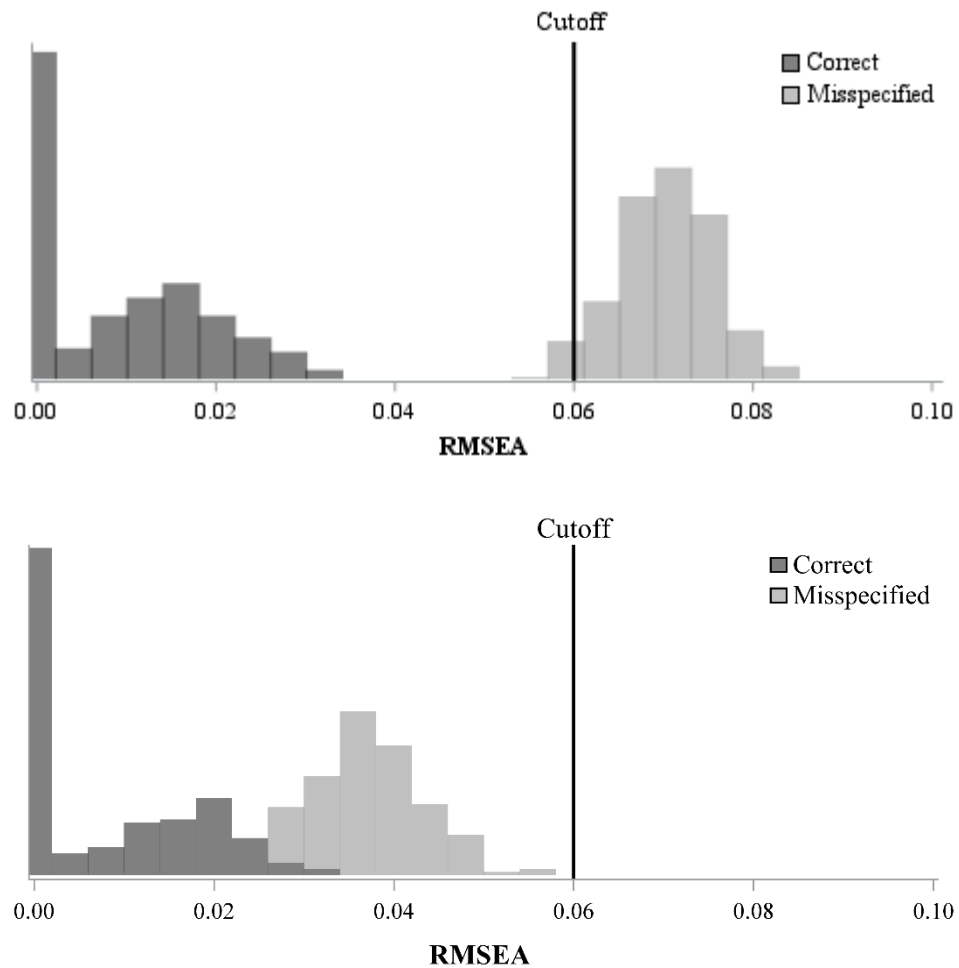


Figure 1. Replication of Hu and Bentler's simulation for deriving an RMSEA cutoff. The top panel replicates their conditions exactly, showing that an RMSEA value of 0.06 differentiates between the distribution for models that are known to be correct (in dark grey) and the distribution for models that are known to be misspecified (in light grey). The bottom panel replicates their conditions with the exception of using standardized loadings between 0.50 and 0.60 (instead of 0.70 and 0.80 as in the original study). As a result, the means and variances of the fit index distributions shift and the 0.06 RMSEA cutoff no longer successfully differentiates between correct and misspecified models.

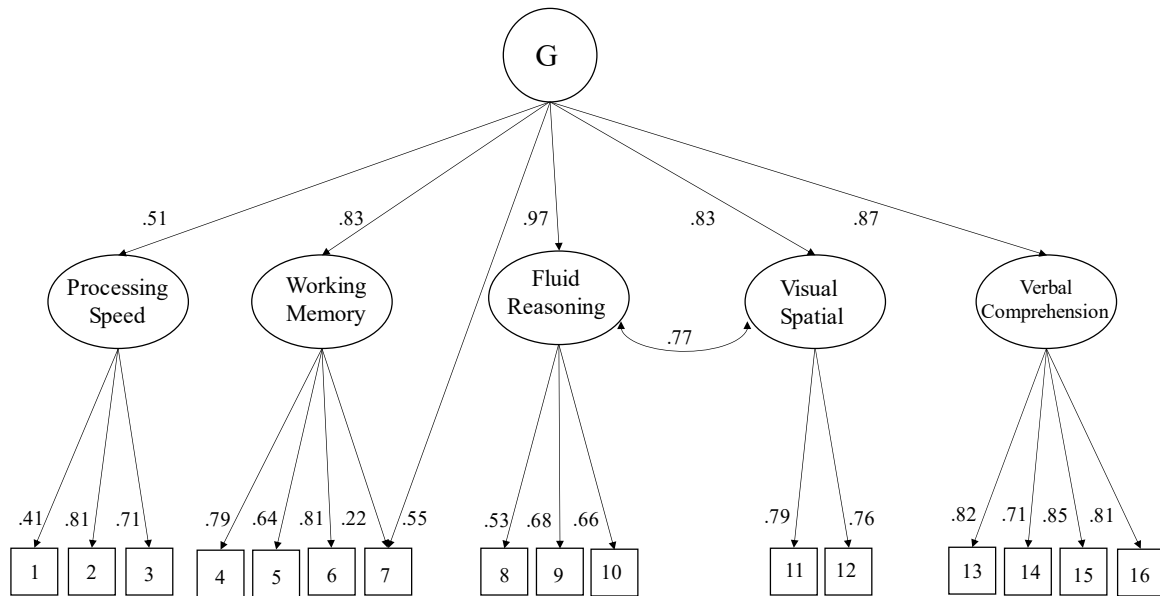


Figure 2. Path diagram for hierarchical factor model from Reynolds & Keith (2017), $N = 2200$. Values next to paths are standardized estimates. Numbers inside the observed variable rectangles are item numbers.

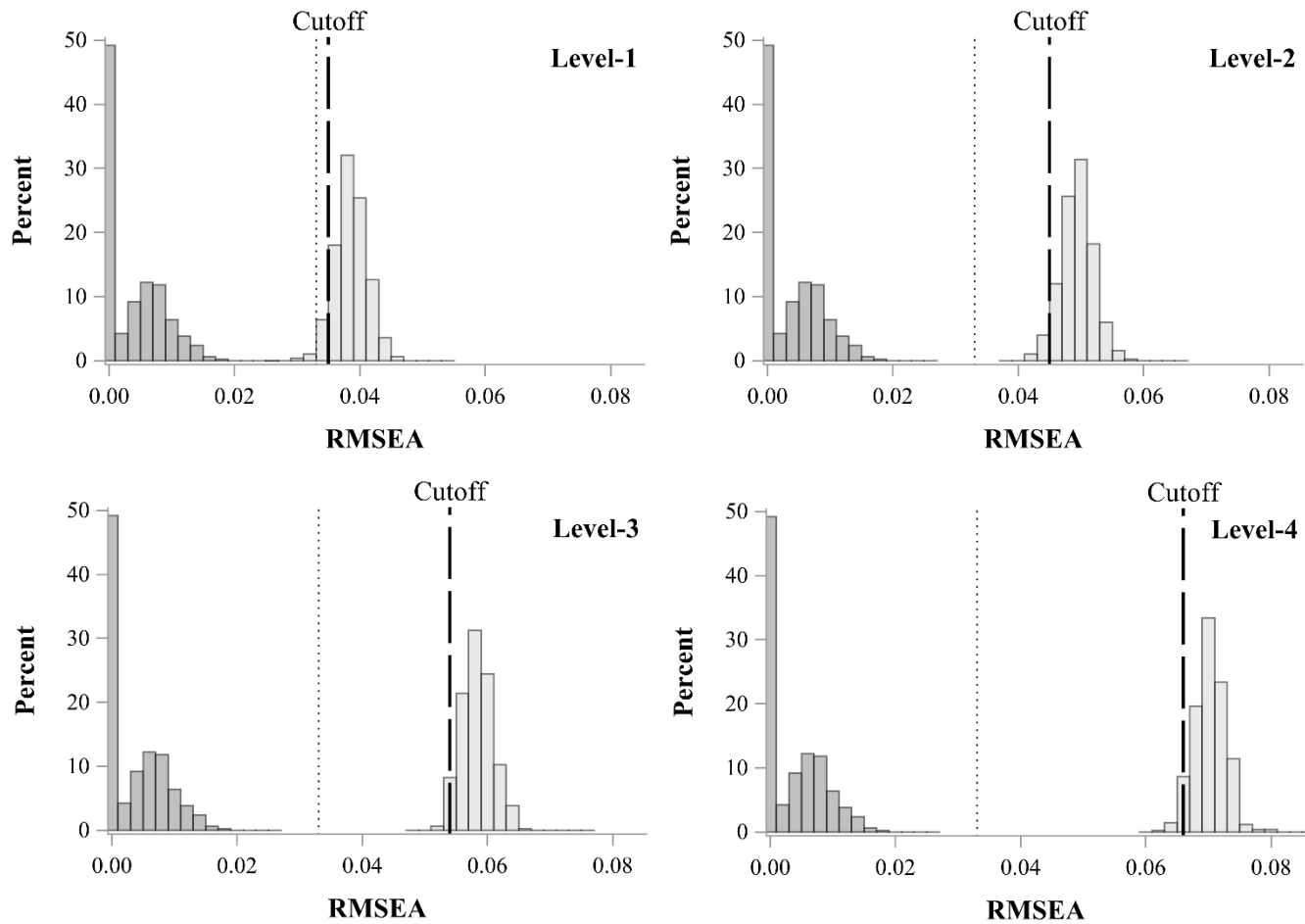


Figure 3. Plots of RMSEA values for testing measurement misspecification for the hierarchical factor model in Reynolds & Keith (2017) for each of the four levels. In each plot, the dark grey distribution represents RMSEA values assuming the fitted model were correct and the light grey distribution represents RMSEA values assuming the fitted model were hypothetically misspecified. The thick dashed line represents the DFI cutoff that would reject at least 95% of the replications in the light grey distribution. The dotted line is the 0.033 RMSEA value from the fitted model in Reynolds & Keith (2017).